

# Coupling of Slack and Na<sub>v</sub>1.6 sensitizes Slack to quinidine blockade and guides anti-seizure strategy development

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Tian Yuan, Yifan Wang, Yuchen Jin, Shuai Xu, Heng Zhang, Qian Chen, Na Li, Xinyue Ma, Huifang Song, Chao Peng, Hui Yang, Ze Geng, Jie Dong, Guifang Duan, Qi Sun, Yang Yang, Fan Yang, Zhuo Huang ✉

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China • NHC and CAMS Key Laboratory of Medical Neurobiology, MOE Frontier Science Center for Brain Research and Brain-Machine Integration, School of Brain Science and Brain Medicine, Zhejiang University, Hangzhou, Zhejiang, 310058, China • Department of Medicinal Chemistry and Molecular Pharmacology, College of Pharmacy, Purdue University, West Lafayette, IN 47907, USA • Department of Biophysics, Kidney Disease Center of the First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, 310058, China • IDG/McGovern Institute for Brain Research, Peking University, Beijing, 100871, China

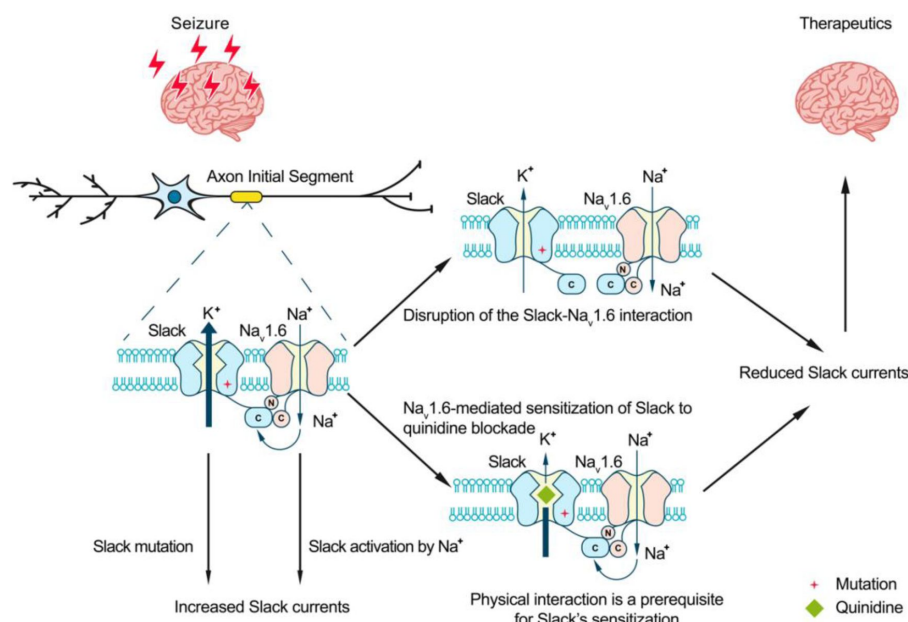
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## Abstract

Quinidine has been used as an anticonvulsant to treat patients with KCNT1-related epilepsy by targeting gain-of-function KCNT1 pathogenic mutant variants. However, the detailed mechanism underlying quinidine's blockade against KCNT1 (Slack) remains elusive. Here, we report a functional and physical coupling of the voltage-gated sodium channel Na<sub>v</sub>1.6 and Slack. Na<sub>v</sub>1.6 binds to and highly sensitizes Slack to quinidine blockade. Homozygous knockout of Na<sub>v</sub>1.6 reduces the sensitivity of native sodium-activated potassium currents to quinidine blockade. Na<sub>v</sub>1.6-mediated sensitization requires the involvement of Na<sub>v</sub>1.6's N- and C-termini binding to Slack's C-terminus, and is enhanced by transient sodium influx through Na<sub>v</sub>1.6. Moreover, disrupting the Slack-Na<sub>v</sub>1.6 interaction by viral expression of Slack's C-terminus can protect against Slack<sup>G269S</sup>-induced seizures in mice. These insights about a Slack-Na<sub>v</sub>1.6 complex challenge the traditional view of "Slack as an isolated target" for anti-epileptic drug discovery efforts, and can guide the development of innovative therapeutic strategies for KCNT1-related epilepsy.

## Graphical abstract



### eLife assessment

The authors report that an interaction between the sodium-activated potassium channel Slack and Nav1.6 sensitizes Slack to inhibition by quinidine. This is an **important** finding because it contributes to our understanding of how the antiseizure drug quinidine affects epilepsy syndromes arising from mutations in the Slack-encoding gene *KCNT1*. The results are largely **compelling**, although additional data would strengthen the claims of a direct channel-channel interaction in vivo.

## Introduction

The sodium-activated potassium ( $K_{Na}$ ) channels Slack and Slick were first identified in guinea pig cardiomyocytes, and were subsequently found to be encoded by two genes of the *Slo2* family<sup>(1),(2)</sup>. Slack channels encoded by the *Slo2.2* (*KCNT1*) gene are gated by  $Na^+$ , while Slick channels encoded by the *Slo2.1* (*KCNT2*) gene are more sensitive to  $Cl^-$  than  $Na^+$ . Slack channels are expressed at high levels in the central nervous system (CNS), especially the cortex and brainstem<sup>(3)–(5)</sup>. Activation of Slack channels by intracellular sodium ions forms delayed outward currents in neurons, contributes to slow afterhyperpolarization (AHP) following repeated action potentials, and modulates the firing frequency of neurons<sup>(6),(7)</sup>.

Mutations in the *KCNT1* gene have been implicated in a wide spectrum of epileptic disorders, including early-onset epilepsy (e.g., epilepsy of infancy with migrating focal seizures (EIMFS), non-EIMFS developmental and epileptic encephalopathies, and autosomal dominant or sporadic sleep-related hypermotor epilepsy (ADSHE))<sup>(8)–(11)</sup>. Over 50 mutations related to seizure disorders have been identified, typically displaying a gain-of-function (GOF) phenotype in heterologous expression systems<sup>(11),(12)</sup>. The prescribed antiarrhythmic drug, quinidine, has emerged as a precision therapy for *KCNT1*-related epilepsy by blocking

Slack mutant variants *in vitro* and conferring decreased seizure frequency and improved psychomotor development in clinical treatment<sup>(13)–(16)</sup>. However, clinical quinidine therapy has shown limited success and contradictory therapeutic effects, probably due to poor blood–brain barrier penetration, dose-limiting off-target effects, phenotype-genotype associations, and rational therapeutic schedule<sup>(16)–(19)</sup>.

Slack requires high intracellular free  $\text{Na}^+$  concentrations ( $[\text{Na}^+]_{\text{in}}$ ) for its activation in neurons ( $K_d$  of  $\sim 66$  mM)<sup>(20)</sup>. However, previous investigations have shown that the  $[\text{Na}^+]_{\text{in}}$  at resting states ( $\sim 10$  mM) is much lower than the  $[\text{Na}^+]_{\text{in}}$  needed for effective Slack activation (e.g.,  $K_d$  value)<sup>(1),(20),(21)</sup>. Therefore, native Slack channels need to localize with  $\text{Na}^+$  sources within a nanodomain to be activated and exert physiological function. Slack channels are known to be functionally coupled with sodium-permeable ion channels in neurons, such as voltage-gated sodium ( $\text{Na}_V$ ) channels and AMPA receptors<sup>(21),(22)</sup>. A question arises as to whether these known  $\text{Na}^+$  sources modulate Slack's sensitivity to quinidine blockade.

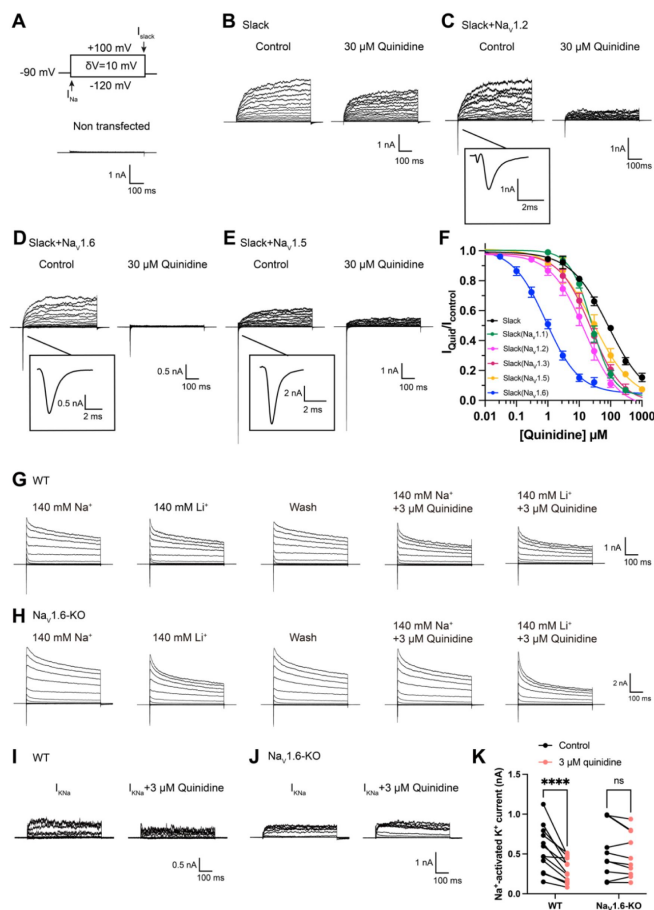
Here, we found that  $\text{Na}_V1.6$  sensitizes Slack to quinidine blockade. Slack and  $\text{Na}_V1.6$  form a complex that functions in  $\text{Na}_V1.6$ -mediated transient sodium influx to sensitize Slack to quinidine blockade in HEK293 cells and in primary cortical neurons. The widespread expression of these channel proteins in cerebral cortex, hippocampus, and cerebellum supports that the  $\text{Na}_V1.6$ -Slack complex is essential for the function of a wide range of electrically excitable neurons, and moreover, that this complex can be viewed as a vulnerable target for drug development to treat KCNT1-related disorders.

## Results

### $\text{Na}_V1.6$ sensitizes Slack to quinidine blockade

Slack currents are activated by sodium entry through voltage-gated sodium ( $\text{Na}_V$ ) channels and ionotropic glutamate receptors (e.g. AMPA receptors)<sup>(21)–(23)</sup>. To investigate potential modulators of Slack's sensitivity to quinidine blockade, we initially focused on the known  $\text{Na}^+$  sources of Slack. Working in HEK293 cells, we co-expressed Slack with AMPA receptor subunits (GluR1, GluR2, GluR3, or GluR4) or  $\text{Na}_V$  channel  $\alpha$  subunits ( $\text{Na}_V1.1$ ,  $\text{Na}_V1.2$ ,  $\text{Na}_V1.3$ , or  $\text{Na}_V1.6$ ), which are highly expressed in the central nervous system<sup>(24)–(27)</sup>. The sensitivity of Slack to quinidine blockade was assessed based on the detected inhibitory effects of quinidine on delayed outward potassium currents<sup>(13),(28)</sup>. Interestingly, all neuronal  $\text{Na}_V$  channels significantly sensitized Slack to 30  $\mu\text{M}$  quinidine blockade, whereas no effect was observed upon co-expression of Slack with GluR1, GluR2, GluR3, or GluR4 (Fig. S1).

When co-expressing Slack with  $\text{Na}_V1.6$  in HEK293 cells,  $\text{Na}_V1.6$  sensitized Slack to quinidine blockade by nearly 100-fold ( $\text{IC}_{50} = 85.13$   $\mu\text{M}$  for Slack expressed alone and an  $\text{IC}_{50} = 0.87$   $\mu\text{M}$  for Slack upon co-expression with  $\text{Na}_V1.6$ ) (Fig. 1A,B,D,F).  $\text{Na}_V1.6$  also exhibited >10-fold selectivity in sensitizing Slack to quinidine blockade against  $\text{Na}_V1.1$ ,  $\text{Na}_V1.2$ , and  $\text{Na}_V1.3$  (Fig. 1C,F, and Supplementary Table 1). When we co-expressed the cardiac sodium channel  $\text{Na}_V1.5$  with Slack, we observed only  $\sim 3$ -fold sensitization of Slack to quinidine blockade (Fig. 1E,F). These results together indicate that the apparent functional coupling between Slack and  $\text{Na}_V$  channels is  $\text{Na}_V$ -channel-subtype-specific, with  $\text{Na}_V1.6$  being particularly impactful in Slack's responsivity to quinidine blockade.



**Figure 1.**

## Na<sub>V</sub>1.6 specifically sensitizes Slack to quinidine blockade.

(A) The voltage protocol and current traces from control (non-transfected) HEK293 cells. The arrows on the voltage protocol indicate the onset of inward sodium currents through Na<sub>V</sub> channels and delayed outward potassium currents through Slack channels. The currents were evoked by applying 600-ms step pulses to voltages varying from −120 mV to +100 mV in 10 mV increments, with a holding potential of −90 mV and a stimulus frequency of 0.20 Hz. (B) Example current traces from HEK293 cells expressing Slack alone. The left traces show the family of control currents; the right traces show Slack currents remaining after application of 1 μM quinidine in the bath solution. (C–E) Example current traces from HEK293 cells co-expressing Slack with Na<sub>V</sub>1.2 (C), Na<sub>V</sub>1.6 (D), or Na<sub>V</sub>1.5 (E) channels before and after application of 30 μM quinidine. (F) The concentration-response curves for blocking of Slack by quinidine at +100 mV upon expression of Slack alone (*n* = 6) and co-expression of Slack with Na<sub>V</sub>1.1 (*n* = 7), Na<sub>V</sub>1.2 (*n* = 10), Na<sub>V</sub>1.3 (*n* = 13), Na<sub>V</sub>1.5 (*n* = 9), or Na<sub>V</sub>1.6 (*n* = 19). Please refer to Supplementary Table 1 for IC<sub>50</sub> values. (G–H) Delayed outward currents in primary cortical neurons from Na<sub>V</sub>1.6 knockout C3HeB/FeJ mice (Na<sub>V</sub>1.6-KO) (H) and the wild-

type littermate controls (WT) (G). Current traces were elicited by 600-ms step pulses to voltages varying from −120 mV to +100 mV in 20 mV increments, with a holding potential of −70 mV, and recorded with different bath solutions in the following order: Na<sup>+</sup>-based bath solution (*I*<sub>Control</sub>), replacement of external Na<sup>+</sup> with Li<sup>+</sup> in equivalent concentration (*I*<sub>Li</sub>), washout of quinidine by Na<sup>+</sup>-based bath solution (*I*<sub>Wash</sub>), Na<sup>+</sup>-based bath solution with 3 μM quinidine (*I*<sub>Quid</sub>), Li<sup>+</sup>-based bath solution with 3 μM quinidine (*I*<sub>Li+Quid</sub>). The removal and subsequent replacement of extracellular Na<sup>+</sup> revealed the *I*<sub>KNa</sub> in neurons. (I–J), The sensitivity of native sodium-activated potassium currents (*I*<sub>KNa</sub>) to 3 μM quinidine blockade in WT (I) and Na<sub>V</sub>1.6-KO (J) neurons. *I*<sub>KNa</sub> before application of quinidine was obtained from the subtraction of *I*<sub>Control</sub> and *I*<sub>Li</sub>. Maintained *I*<sub>KNa</sub> after application of 3 μM quinidine was obtained from the subtraction of *I*<sub>Quid</sub> and *I*<sub>Li+Quid</sub>. (K) Summarized amplitudes of *I*<sub>KNa</sub> before and after application of 3 μM quinidine in WT (black, *n* = 12) and Na<sub>V</sub>1.6-KO (red, *n* = 10) primary cortical neurons. \*\*\*\**p* < 0.0001, Two-way repeated measures ANOVA followed by Bonferroni's post hoc test.

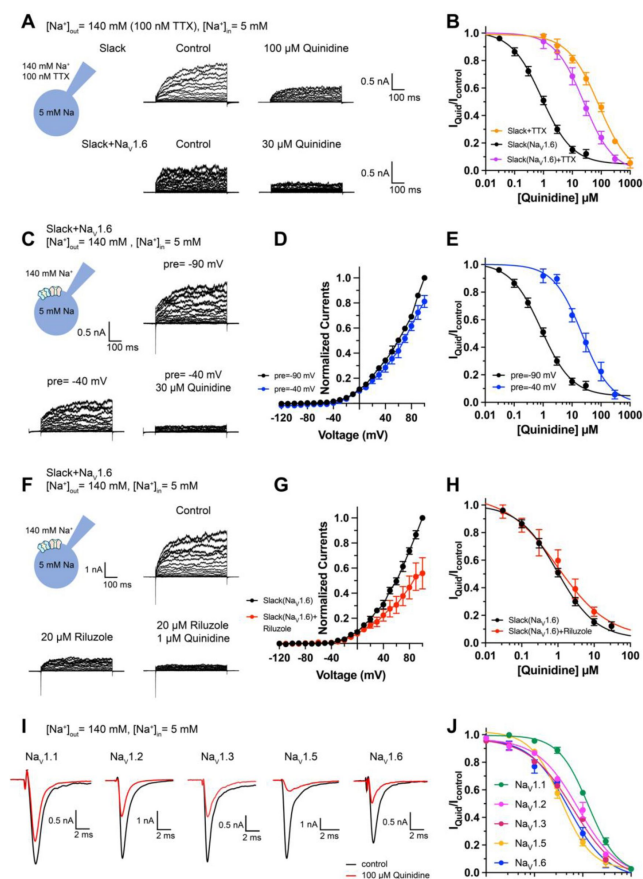
Slack and Slick are both K<sub>Na</sub> channels (with 74% sequence identity) and adopt similar structures<sup>(29)</sup>, and Slick is also blocked by quinidine<sup>(30)</sup>. We next assessed whether Na<sub>V</sub>1.6 sensitizes Slick to quinidine blockade and observed that, similar to Slack, co-expression of Slick and Na<sub>V</sub>1.6 in HEK293 cells resulted in a sensitization of Slick to quinidine blockade (7-fold) (Fig. S2A,B). These results support that Na<sub>V</sub>1.6 regulates both Slack and Slick and that Na<sub>V</sub>1.6 can sensitize K<sub>Na</sub> channels to quinidine blockade *in vitro*.

We also asked whether Na<sub>V</sub>1.6 sensitizes native K<sub>Na</sub> channels to quinidine blockade *in vivo*. We performed whole-cell patch-clamp recordings in primary cortical neurons from Na<sub>V</sub>1.6-knockout C3HeB/FeJ mice and the wild-type littermate controls. K<sub>Na</sub> currents (*I*<sub>KNa</sub>) were

isolated by replacing sodium ions with equivalent lithium ions in the bath solution (Fig. 1G,H). 3  $\mu\text{M}$  quinidine significantly blocked native  $I_{\text{KNa}}$  (44%) in wild-type neurons (Fig. 1I,K), while the same concentration of quinidine had no significant effect on  $I_{\text{KNa}}$  in  $\text{Na}_v1.6$ -knockout ( $\text{Na}_v1.6\text{-KO}$ ) neurons (Fig. 1J,K). These results support that  $\text{Na}_v1.6$  is required for the observed high sensitivity of native  $\text{K}_{\text{Na}}$  channels to quinidine blockade.

## Transient sodium influx through $\text{Na}_v1.6$ enhances $\text{Na}_v1.6$ -mediated sensitization of Slack to quinidine blockade

We next investigated the biomolecular mechanism underlying  $\text{Na}_v1.6$ -mediated sensitization of Slack to quinidine blockade. Considering that Slack currents are activated by sodium influx<sup>(22),(29)</sup>, we initially assessed the effects of  $\text{Na}_v1.6$ -mediated sodium influx on sensitizing Slack to quinidine blockade. We used 100 nM tetrodotoxin (TTX) to block  $\text{Na}_v1.6$ -mediated sodium influx (Fig. S3A,B)<sup>(31)</sup>. In HEK293 cells expressing Slack alone, 100 nM TTX did not affect Slack currents; nor did it affect Slack's sensitivity to quinidine blockade ( $\text{IC}_{50} = 83.27 \mu\text{M}$ ) (Fig. 2A,B and Fig. S3C,D). In contrast, upon co-expression of Slack and  $\text{Na}_v1.6$  in HEK293 cells, bath-application of 100nM TTX significantly reduced the effects of  $\text{Na}_v1.6$  in sensitizing Slack to quinidine blockade ( $\text{IC}_{50} = 25.04 \mu\text{M}$ ) (Fig. 2A,B). These findings support that sodium influx through  $\text{Na}_v1.6$  contributes to  $\text{Na}_v1.6$ -mediated sensitization of Slack to quinidine blockade.



**Figure 2.**

### Blocking transient sodium influx through $\text{Na}_v1.6$ reduces $\text{Na}_v1.6$ -mediated sensitization of Slack to quinidine blockade.

(A) Example current traces from HEK293 cells expressing Slack alone (top) and co-expressing Slack with  $\text{Na}_v1.6$  (bottom), with 100nM TTX in the bath solution. The left traces show the family of control currents; the right traces show Slack currents remaining after application of quinidine. (B) The concentration-response curves for blocking of Slack by quinidine upon expression of Slack alone ( $n = 3$ ) and co-expression of Slack with  $\text{Na}_v1.6$  ( $n = 7$ ), with 100 nM TTX in the bath solution. (C) Top, example current traces recorded from a HEK293 cell co-expressing Slack with  $\text{Na}_v1.6$  and evoked from a 100-ms prepulse (pre) of  $-90 \text{ mV}$ . Bottom, example current traces recorded from the same cell but evoked from a 100-ms prepulse of  $-40 \text{ mV}$ , before and after application of quinidine. (D) I-V curves of Slack upon co-expression with  $\text{Na}_v1.6$ . The currents were evoked from a prepulse of  $-90 \text{ mV}$  (black) or  $-40 \text{ mV}$  (blue). (E) The concentration-response curves for blocking of Slack by quinidine with a prepulse of  $-90 \text{ mV}$  (black,  $n = 19$ ) or  $-40 \text{ mV}$  (blue,  $n = 5$ ). (F) Top, example current traces recorded from a HEK293 cell co-expressing Slack and  $\text{Na}_v1.6$  without riluzole in the bath solution.

Bottom, example current traces recorded from the same cell with 20  $\mu\text{M}$  riluzole in the bath solution, before and after application of quinidine. (G) I-V curves of Slack upon co-expression with  $\text{Na}_v1.6$  before (black) and after (red) application of



20  $\mu$ M riluzole into bath solution. The concentration-response curves for blocking of Slack by quinidine upon co-expression of Slack with Na<sub>v</sub>1.6, without ( $n = 19$ ) or with ( $n = 6$ ) 20  $\mu$ M riluzole in the bath solution. (I–J) The sensitivity of Na<sub>v</sub> channel subtypes to quinidine blockade upon expression of Na<sub>v</sub> alone in HEK293 cells. Example current traces (I) were evoked by a 50-ms step depolarization to 0 mV from a holding potential of  $-90$  mV. The Concentration-response curves for blocking of Na<sub>v</sub> channel subtypes by quinidine (J) were shown on the right panel ( $n = 5$  for Na<sub>v</sub>1.1,  $n = 3$  for Na<sub>v</sub>1.2,  $n = 6$  for Na<sub>v</sub>1.3,  $n = 6$  for Na<sub>v</sub>1.5, and  $n = 4$  for Na<sub>v</sub>1.6).

It is known that Na<sub>v</sub>1.6-mediated sodium influx involves a transient inward flux that reaches a peak before subsequently decaying to the baseline within a few milliseconds; this is termed a transient sodium current ( $I_{NaT}$ )<sup>(32)</sup>. A small fraction of Na<sub>v</sub>1.6 currents are known to persist after the rapid decay of  $I_{NaT}$ , and these are termed persistent sodium currents ( $I_{NaP}$ )<sup>(33)</sup>. We isolated  $I_{NaT}$  and  $I_{NaP}$  to explore their potential contributions in sensitizing Slack to quinidine blockade. We selectively inactivated  $I_{NaT}$  using a depolarized prepulse of  $-40$  mV (Fig. S4A) and selectively blocked  $I_{NaP}$  by bath- application of 20  $\mu$ M riluzole, which is a relatively specific  $I_{NaP}$  blocker that is known to stabilize inactivated-state Na<sub>v</sub> channels and delay recovery from inactivation<sup>(34),(35)</sup>. Our findings ultimately confirmed that the 20  $\mu$ M riluzole selectively blocked  $I_{NaP}$  compared to  $I_{NaT}$  in HEK293 cells co-expressing Slack and Na<sub>v</sub>1.6 (Fig. S4B) and that 20  $\mu$ M riluzole had no effect on Slack currents when expressed alone (Fig. S4C,D).

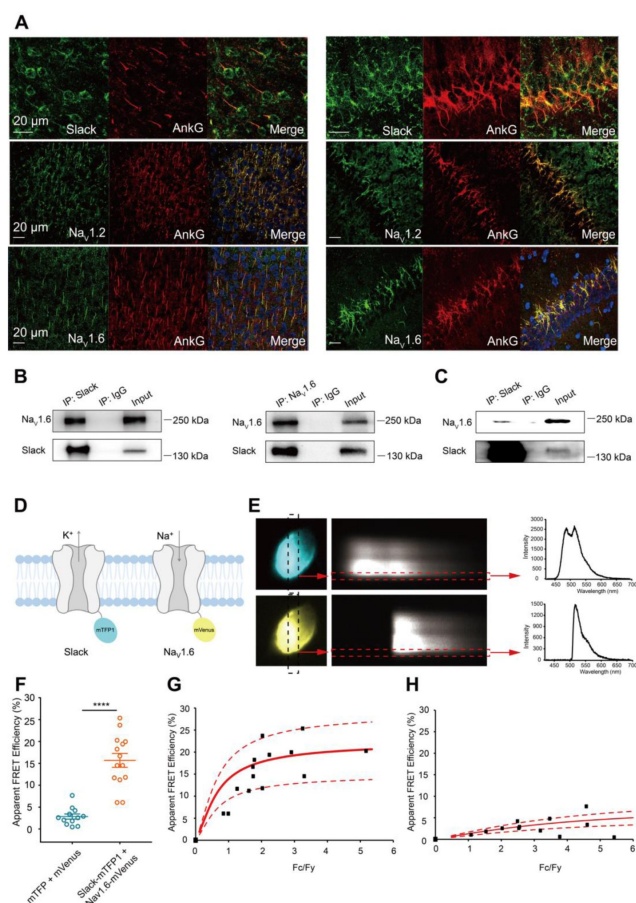
Consistent with previous investigations<sup>(21),(22)</sup>, inactivating  $I_{NaT}$  reduced whole-cell Slack currents by 20%, and blocking  $I_{NaP}$  reduced Slack currents by 40% (Fig. 2C,D,F,G), supporting that Na<sub>v</sub>1.6-mediated sodium influx activates Slack. Interestingly, inactivating  $I_{NaT}$  resulted in a  $> 20$ -fold decrease in Slack's sensitization to quinidine blockade ( $IC_{50} = 22.26$   $\mu$ M) (Fig. 2C,E). In contrast, blocking  $I_{NaP}$  had no effect on Slack's sensitization to quinidine blockade ( $IC_{50} = 1.60$   $\mu$ M) (Fig. 2F,H). These findings indicate that Na<sub>v</sub>1.6 sensitizes Slack to quinidine blockade via  $I_{NaT}$  but not  $I_{NaP}$ .

Given that Slack current amplitudes are sensitive to sodium influx, and considering that quinidine is a sodium channel blocker, we examined whether Na<sub>v</sub>1.6 has higher sensitivity to quinidine blockade than other Na<sub>v</sub> channel subtypes, which could plausibly explain the observed increased strength of sensitization. We used whole-cell patch-clamping to assess the sensitivity of Na<sub>v</sub>1.1, Na<sub>v</sub>1.2, Na<sub>v</sub>1.3, Na<sub>v</sub>1.5, and Na<sub>v</sub>1.6 to quinidine blockade. These sodium channels exhibited similar levels of quinidine sensitivity ( $IC_{50}$  values in the range of 35.61–129.84  $\mu$ M) (Fig. 2I,J and Supplementary Table 3), all of which were at least 40-fold lower than the Na<sub>v</sub>1.6-mediated sensitization of Slack to quinidine blockade (Fig. 1F). Additionally, co-expressing Slack with Na<sub>v</sub>1.1, Na<sub>v</sub>1.2, Na<sub>v</sub>1.3, Na<sub>v</sub>1.5, or Na<sub>v</sub>1.6 in HEK293 cells did not change the sensitivity of these Na<sub>v</sub> channel subtypes to quinidine blockade (Fig. S5 and Supplementary Table 3). Thus, differential quinidine affinity for specific Na<sub>v</sub> channel subtypes cannot explain the large observed Na<sub>v</sub>1.6-mediated sensitization of Slack to quinidine blockade. Moreover, it is clear that Na<sub>v</sub>1.6-mediated sensitization of Slack to quinidine blockade is directly mediated by  $I_{NaT}$ , rather than through some secondary effects related to Na<sub>v</sub>1.6's higher sensitivity to quinidine blockade.

## Slack interacts with Na<sub>v</sub>1.6 *in vitro* and *in vivo*

We found that the specific voltage-gated sodium channel blocker TTX did not completely abolish the effects of Na<sub>v</sub>1.6 on sensitizing Slack to quinidine blockade (Fig. 2B), so it appears that a sodium-influx-independent mechanism is involved in the observed Na<sub>v</sub>1.6-mediated sensitization of Slack to quinidine blockade. We therefore investigated a potential physical interaction between Slack and Na<sub>v</sub>1.6. We initially assessed the cellular distribution of

Na<sub>v</sub>1.2, Na<sub>v</sub>1.6, and Slack in the hippocampus and the neocortex of mouse. Consistent with previous studies<sup>(36),(37)</sup>, Na<sub>v</sub>1.2 and Na<sub>v</sub>1.6 were localized to the axonal initial segment (AIS) of neurons, evident as the co-localization of Na<sub>v</sub> and AnkG, a sodium channel-associated protein known to accumulate at the AIS (Fig. 3A). Slack channels were also localized to the AIS of these neurons (Fig. 3A), indicating that Slack channels are located in close proximity to Na<sub>v</sub>1.6 channels, and supporting their possible interaction *in vivo*. Moreover, Na<sub>v</sub>1.6 was co-immunoprecipitated with Slack in homogenates from mouse cortical and hippocampal tissues and from HEK293T cells co-transfected with Slack and Na<sub>v</sub>1.6 (Fig. 3B,C), supporting that a physical interaction between Slack and Na<sub>v</sub>1.6 occurs *in vivo*.



**Figure 3.**

### Slack physically interacts with Na<sub>v</sub>1.6 *in vitro* and *in vivo*.

(A) Immunofluorescence of Slack, Na<sub>v</sub>1.2, Na<sub>v</sub>1.6 (green), and AnkG (red) in hippocampus CA1 (left) and neocortex (right). Confocal microscopy images were obtained from Coronal brain slices of C57BL/6 mice. (B) Coimmunoprecipitation (Co-IP) of Slack and Na<sub>v</sub>1.6 in cell lysates from HEK293T cells co-transfected with Slack and Na<sub>v</sub>1.6. (C) Co-IP of Slack and Na<sub>v</sub>1.6 in mouse brain tissue lysates. Input volume corresponds to 10% of the total lysates for Co-IP. (D) A schematic diagram showing the fluorescence-labeled Slack and Na<sub>v</sub>1.6. mTFP1 and mVenus were fused to the C-terminal region of Slack (Slack-mTFP1) and Na<sub>v</sub>1.6 (Na<sub>v</sub>1.6-mVenus), respectively. (E) FRET imaging of Slack-mTFP1 and Na<sub>v</sub>1.6-mVenus co-expressed in HEK293 cells. The emission spectra measured from the edge of cell (dotted arrows in red) are used for FRET efficiency calculation. (F) The apparent FRET efficiency measured from cells co-expressing the fluorophore-tagged ion channels or co-expressing the fluorophores. \*\*\*\*  $p < 0.0001$ , unpaired two-tailed Student's *t* test. (G-H) The FRET efficiency measured from cells co-expressing the fluorophore-tagged ion channels (G), or from cells co-expressing fluorophores (H). The efficiency value was plotted as a function of the fluorescence

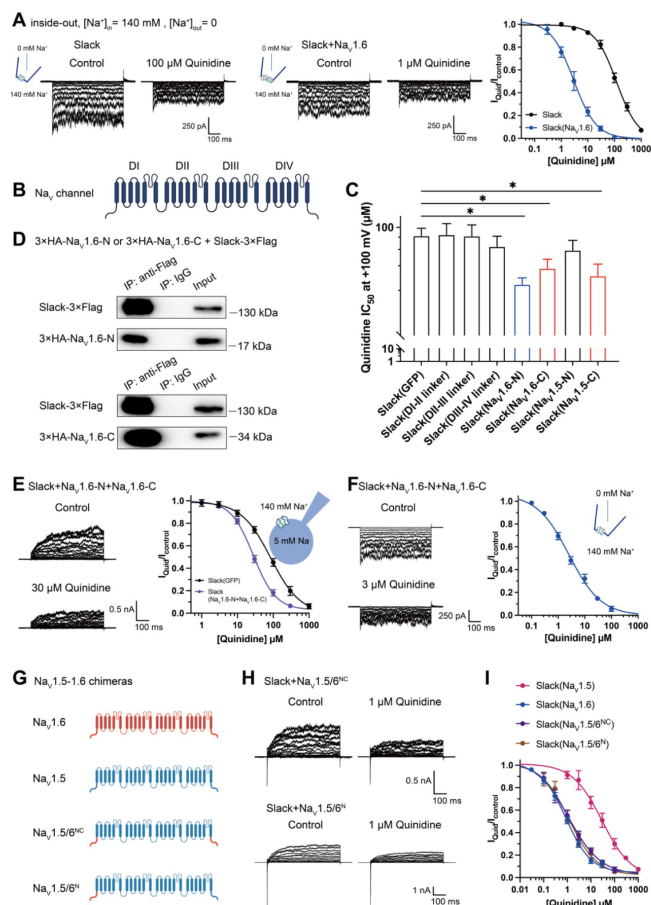
intensity ratio between mTFP1 and mVenus ( $F_c/F_y$ ). Each symbol represents a single cell. The solid curve represents the FRET model that yields the best fit; dotted curves represent models with 5% higher or lower FRET efficiencies.

To assess the interaction between Slack and Na<sub>v</sub>1.6 inside living cells, we performed a FRET assay in transfected HEK293T cells<sup>(38)</sup>. Briefly, we genetically fused mTFP1 and mVenus to the C-terminal regions of Slack and Na<sub>v</sub>1.6, respectively (Fig. 3D). Upon imaging the emission spectra cells co-expressing Na<sub>v</sub>1.6-mVenus and Slack-mTFP1 (measured at the plasma membrane region) (Fig. 3E), we detected positive FRET signals, indicating a Slack-Na<sub>v</sub>1.6 interaction (Fig. 3F,H). The plasma membrane regions from HEK293T cells co-transfected with Na<sub>v</sub>1.6 and Slack showed FRET efficiency values much larger than a negative control (in which standalone mVenus and mTFP1 proteins were co-expressed) (Fig. 3G,H), indicating that Slack channels reside in close spatial proximity (less than 10 nm) to Na<sub>v</sub>1.6 channels in membranes of living cells.

We next characterized the consequences of the Slack-Nav<sub>1.6</sub> interaction in HEK293 cells using whole-cell recordings. Slack increased the rate of recovery from fast inactivation of Nav<sub>1.6</sub> (Fig. S6E), with no significant effects on the steady-state activation, steady-state fast inactivation, or ramp currents (Fig. S6C,D,F and Supplementary Table 3). Additionally, we found that Nav<sub>1.6</sub> had no significant effects on the activation rate or the current-voltage (I-V) relationship of Slack currents (Fig. S6A,B). These results indicate that the physical interaction between Slack and Nav<sub>1.6</sub> produces functional consequences. Taken together, these findings support functional and physical coupling of Slack and Nav<sub>1.6</sub> *in vitro* and *in vivo*.

## Nav<sub>1.6</sub>'s N- and C-termini bind to Slack's C-terminus and sensitize Slack to quinidine blockade

To explore whether the physical interaction between Slack and Nav<sub>1.6</sub> is required for sodium-influx-mediated sensitization of Slack to quinidine blockade, we performed inside-out patch-clamp recordings on HEK293 cells transfected with Slack alone or co-transfected with Slack and Nav<sub>1.6</sub>. Note that in these experiments the intracellular sodium concentration ( $[Na^+]_{in}$ ) was raised to 140 mM (a concentration at which most Slack channels can be activated<sup>(20)</sup>), seeking to mimic the increased intracellular sodium concentration upon sodium influx. When expressing Slack alone, increasing the sodium concentration did not sensitize Slack to quinidine blockade ( $IC_{50} = 120.42 \mu M$ ) (Fig. 4a). However, upon co-expression of Nav<sub>1.6</sub> and Slack, Nav<sub>1.6</sub> significantly sensitized Slack to quinidine blockade ( $IC_{50} = 2.91 \mu M$ ) (Fig. 4A). These results support that physical interaction between Slack and Nav<sub>1.6</sub> is a prerequisite for sodium-influx-mediated sensitization of Slack to quinidine blockade.



**Figure 4.**

### Nav<sub>1.6</sub>'s N- and C-termini interacting with Slack is a prerequisite for Nav<sub>1.6</sub>-mediated sensitization of Slack to quinidine blockade.

(A) The sensitivity of Slack to quinidine blockade upon expression of Slack alone ( $n = 3$ ) and co-expression of Slack with Nav<sub>1.6</sub> ( $n = 3$ ) from excised inside-out patches. The pipette solution contained (in mM) 130 KCl, 1 EDTA, 10 HEPES and 2  $MgCl_2$  (pH 7.3); the bath solution contained (in mM) 140 NaCl, 1 EDTA, 10 HEPES and 2  $MgCl_2$  (pH 7.4). The membrane voltage was held at 0 mV and stepped to voltages varying from -100 mV to 0 mV in 10 mV increments. Example current traces were shown on the left panel. The concentration-response curves for blocking of Slack by quinidine were shown on the right panel. (B) Domain architecture of the human Nav<sub>1.6</sub> channel pore-forming  $\alpha$  subunit. (C) Calculated  $IC_{50}$  values at +100 mV of quinidine on Slack upon co-expression with indicated cytoplasmic fragments from Nav<sub>1.6</sub> channels. For Nav<sub>1.6</sub>, cytoplasmic fragments used include N-terminus (Nav<sub>1.6</sub>-N, residues 1-132), inter domain linkers (Domain I-II linker, residues 409-753; Domain II-III linker, residues 977-1199; Domain III-IV linker, residues



1461-1523), and C-terminus (Na<sub>v</sub>1.6-C, residues 1766-1980). For Na<sub>v</sub>1.5, cytoplasmic fragments used include N-terminus (Na<sub>v</sub>1.5-N, residues 1-131) and C-terminus (Na<sub>v</sub>1.5-C, residues 1772-2016). **(D)** Co-IP of Slack and terminal domains of Na<sub>v</sub>1.6 in cell lysates from HEK293T cells co-expressing 3×Flag-tagged Slack (Slack-3×Flag) and 3×HA-tagged termini of Na<sub>v</sub>1.6 (3×HA-Na<sub>v</sub>1.6-N or 3×HA-Na<sub>v</sub>1.6-C). The 3×Flag tag was fused to the C-terminal region of Slack and the 3×HA tag was fused to the N-terminal region of Na<sub>v</sub>1.6's fragments. **(E)** The sensitivity of Slack to quinidine blockade upon co-expression of Slack with GFP (*n* = 12) or N- and C-termini of Na<sub>v</sub>1.6 (*n* = 11), from whole-cell recordings. **(F)** The sensitivity of Slack to quinidine blockade upon co-expression of Slack with N- and C-termini of Na<sub>v</sub>1.6, from excised inside-out recordings (*n* = 10, using the same protocols as in Fig. 4A). Example current traces before and after application of quinidine were shown on the left panel. The concentration-response curves were shown on the right panel. **(G)** A schematic diagram of the Na<sub>v</sub>1.5-1.6 chimeric channels (Na<sub>v</sub>1.5/6<sup>NC</sup> and Na<sub>v</sub>1.5/6<sup>N</sup>) used in this study. **(H)** Example current traces recorded from HEK293 cells co-expressing Slack and Na<sub>v</sub>1.5-1.6 chimeras before and after application of the indicated concentration of quinidine. **(I)** The concentration-response curves for blocking of Slack by quinidine upon co-expression of Slack with Na<sub>v</sub>1.5 (*n* = 9), Na<sub>v</sub>1.6 (*n* = 19), Na<sub>v</sub>1.5/6<sup>NC</sup> (*n* = 9), or Na<sub>v</sub>1.5/6<sup>N</sup> (*n* = 9).

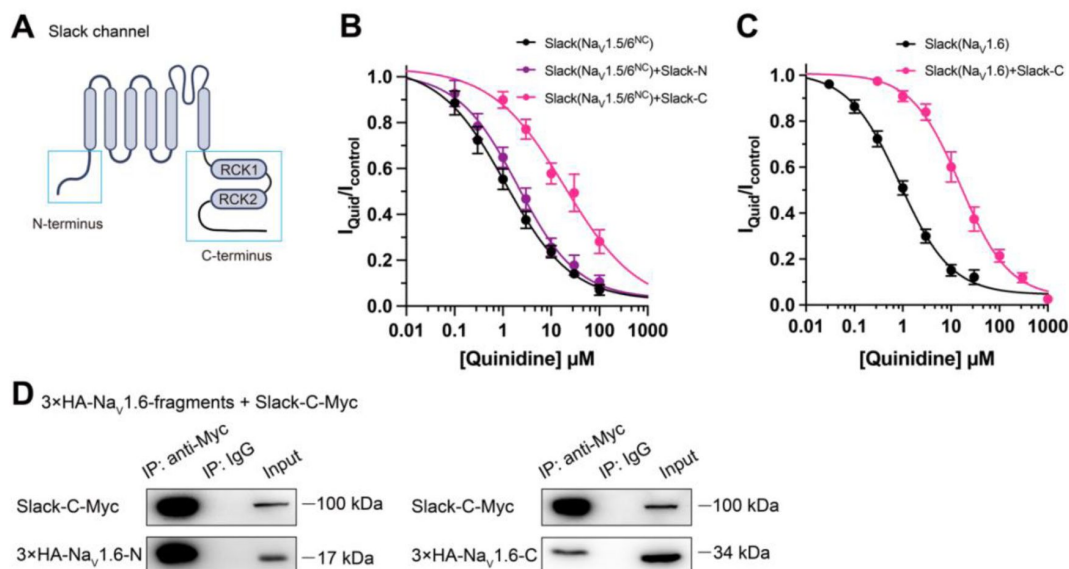
To investigate which interacting domains mediate Na<sub>v</sub>1.6's sensitization of Slack to quinidine blockade, we focused on Na<sub>v</sub>1.6's cytoplasmic fragments, including its N-terminus, inter domain linkers, and C-terminus (Fig. 4B). Whole-cell recordings from HEK293 cells co-expressing Slack with these Na<sub>v</sub>1.6 fragments revealed that the N-terminus and C-terminus of Na<sub>v</sub>1.6 significantly enhanced the sensitivity of Slack to quinidine blockade (IC<sub>50</sub> = 31.59 μM for Slack upon co-expression with Na<sub>v</sub>1.6's N-terminus and IC<sub>50</sub> = 43.70 μM for Slack upon co-expression with Na<sub>v</sub>1.6's C-terminus); note that the inter domain linkers had no effect (Fig. 4C).

Subsequent co-immunoprecipitation and glutathione S-transferase (GST) pull down assays of HEK293T cell lysates experimentally confirmed that Na<sub>v</sub>1.6's N- and C-termini each interact with Slack (Fig. 4D and Fig. S7). Additionally, whole-cell recordings using an [Na<sup>+</sup>]<sub>in</sub> of 5 mM again showed that Na<sub>v</sub>1.6's N-terminus and Na<sub>v</sub>1.6's C-terminus sensitize Slack to quinidine blockade (IC<sub>50</sub> = 27.87 μM) (Fig. 4E). And inside-out recordings using an [Na<sup>+</sup>]<sub>in</sub> of 140 mM showed that co-expression of Slack, Na<sub>v</sub>1.6's N-terminus, and Na<sub>v</sub>1.6's C-terminus resulted in obvious sensitization of Slack to quinidine blockade, with an IC<sub>50</sub> of 2.57 μM (Fig. 4F)—thus fully mimicking the aforementioned effects of full-length Na<sub>v</sub>1.6 (IC<sub>50</sub> = 2.91 μM) (Fig. 4A). These findings support that the binding of Na<sub>v</sub>1.6's N- and C-termini to Slack is required for Na<sub>v</sub>1.6's sensitization of Slack to quinidine blockade.

Recalling that Na<sub>v</sub>1.5 had the least pronounced effect in sensitizing Slack to quinidine blockade among all examined Na<sub>v</sub> channels (IC<sub>50</sub> = 29.46 μM) (Fig. 1F and Supplementary Table 1). We constructed Na<sub>v</sub>1.5-1.6 chimeras to test the roles of Na<sub>v</sub>1.6's N- and C-termini in sensitizing Slack to quinidine blockade. The replacement of both the N- and C-termini of Na<sub>v</sub>1.5 with Na<sub>v</sub>1.6's N- and C-termini (namely Na<sub>v</sub>1.5/6<sup>NC</sup>) fully mimicked effects of Na<sub>v</sub>1.6 in sensitizing Slack to quinidine blockade (IC<sub>50</sub> = 1.13 μM) (Fig. 4G-I). We also found that replacement of Na<sub>v</sub>1.5's N-terminus with Na<sub>v</sub>1.6's N-terminus (namely Na<sub>v</sub>1.5/6<sup>N</sup>) fully mimicked the effects of Na<sub>v</sub>1.6 (IC<sub>50</sub> = 1.18 μM) (Fig. 4G-I). Consistently, Na<sub>v</sub>1.5's C-terminus sensitized Slack to quinidine blockade (IC<sub>50</sub> = 37.59 μM), whereas Na<sub>v</sub>1.5's N-terminus had no effect on Slack sensitization (Fig. 4C). These findings support that Na<sub>v</sub>1.6's N-terminus is essential for sensitizing Slack to quinidine blockade.

Having demonstrated that Na<sub>v</sub>1.6 sensitizes Slack via Na<sub>v</sub>1.6's cytoplasmic N- and C-termini, we investigated which domains of Slack interact with Na<sub>v</sub>1.6 and focused on Slack's cytoplasmic fragments, including Slack's N-terminus and C-terminus (Fig. 5A). In HEK293 cells co-expressing Na<sub>v</sub> and Slack, Na<sub>v</sub>-mediated sensitization of Slack to quinidine blockade was significantly attenuated upon the additional expression of Slack's C-terminus, but not of

Slack's N-terminus (Fig. 5B,C), suggesting that Slack's C-terminus can disrupt the Slack-Nav1.6 interaction by competing with Slack for binding to Nav1.6. Consistently, Slack's C-terminus co-immunoprecipitated with Nav1.6's N- and C-termini in HEK293T cell lysates (Fig. 5D). Together, these results support that Slack's C-terminus physically interacts with Nav1.6 and that this interaction is required for Nav1.6's sensitization of Slack to quinidine blockade.



**Figure 5.**

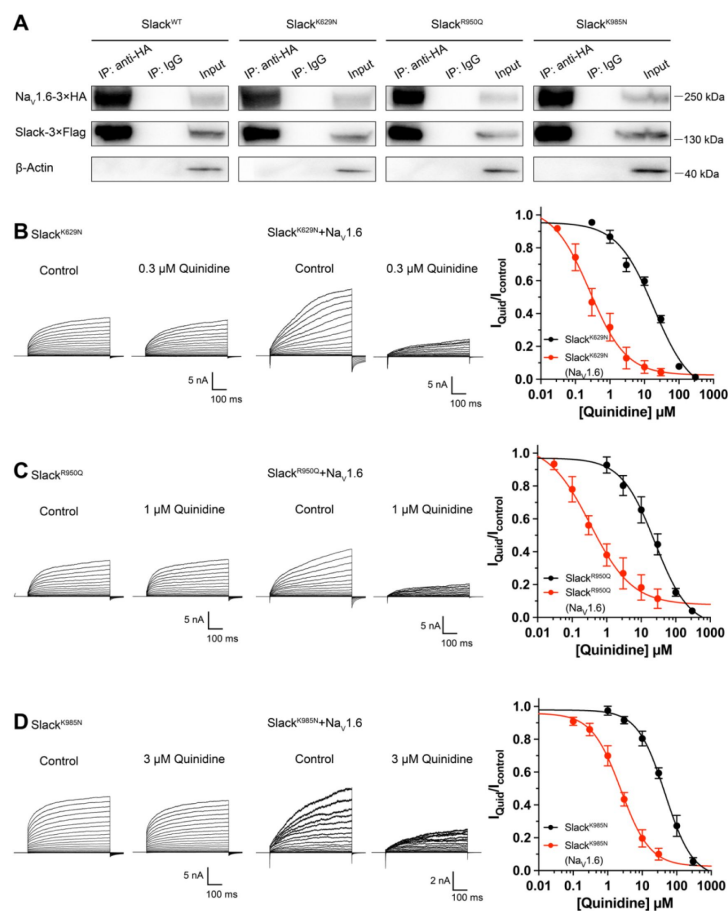
## Slack's C-terminus is required for Nav1.6-mediated sensitization of Slack to quinidine blockade.

(A) Domain architecture of the human Slack channel subunit. Slack's N-terminus (Slack-N, residues 1-116) and C-terminus (Slack-C, residues 345-1235) were shown in the blue boxes. (B) The concentration-response curves for blocking of Slack by quinidine upon additional expression of Slack's N- or C-terminus in HEK293T cells co-expressing Slack and Nav1.5/6<sup>NC</sup>. (C) The concentration-response curves for blocking Slack by quinidine upon additional expression of Slack's C-terminus in HEK293 cells co-expressing Slack and Nav1.6. (D) Co-IP of Myc-tagged Slack's C-terminus (Slack-C-Myc) with 3xHA-tagged Nav1.6's termini (3xHA-Nav1.6-N or 3xHA-Nav1.6-C) in HEK293T cell lysates. The 3xHA tag was fused to the N-terminal region of Nav1.6's fragments, and the Myc tag was fused to the C-terminal region of Slack's fragment.

## Nav1.6 binds to and sensitizes epilepsy-related Slack mutant variants to quinidine blockade

Over 50 mutations in KCNT1 (Slack) have been identified to related to seizure disorders<sup>(11)</sup>. Having established that Nav1.6 can sensitize wild-type Slack to quinidine blockade, we next investigated whether Nav1.6 also sensitizes epilepsy-related Slack mutant variants to quinidine blockade. We chose 3 Slack pathogenic mutant variants (K629N, R950Q, and K985N) initially detected in patients with KCNT1-related epilepsy<sup>(15),(39),(40)</sup>. Considering that these 3 mutations are located in Slack's C-terminus, and recalling that Slack's C-terminus interacts with Nav1.6 (Fig. 5D), we first used co-immunoprecipitation assays and successfully

confirmed that each of these Slack mutant variants interacts with Na<sub>v</sub>1.6 in HEK293T cell lysates (Fig. 6A).



**Figure 6.**

## Na<sub>v</sub>1.6 sensitizes epilepsy-related Slack mutant variants to quinidine blockade.

(A) Co-IP of 3×Flag-tagged Slack or its mutations (Slack-3×Flag) with 3×HA-tagged Na<sub>v</sub>1.6 (Na<sub>v</sub>1.6-3×HA) in HEK293T cell lysates. The tags were all fused to the C-terminal region of wild-type or mutant ion channels. (B–D) The sensitivity of Slack mutant variants (Slack<sup>K629N</sup> [B], Slack<sup>R950Q</sup> [C], and Slack<sup>K985N</sup> [D]) to quinidine blockade upon expression of Slack mutant variants alone and co-expression of Slack mutant variants with Na<sub>v</sub>1.6, before and after application of the indicated concentrations of quinidine. Right, the concentration-response curves for blocking of Slack mutant variants by quinidine upon expression of Slack mutant variants alone (n = 8 for Slack<sup>K629N</sup>, n = 7 for Slack<sup>R950Q</sup>, and n = 5 for Slack<sup>K985N</sup>) and co-expression of Slack mutant variants with Na<sub>v</sub>1.6 (n = 8 for Slack<sup>K629N</sup> upon co-expression with Na<sub>v</sub>1.6, n = 5 for Slack<sup>R950Q</sup> upon co-expression with Na<sub>v</sub>1.6, and n = 7 for Slack<sup>K985N</sup> upon

co-expression with Na<sub>v</sub>1.6). Please refer to Supplementary Table 4 for IC<sub>50</sub> values.

Subsequently, whole-cell recordings revealed that Na<sub>v</sub>1.6 significantly sensitized all of the examined Slack mutant variants to quinidine blockade, with IC<sub>50</sub> values ranging from 0.26 to 2.41 μM (Fig. 6B–D and Supplementary Table 4). These results support that Na<sub>v</sub>1.6 interacts with examined Slack mutant variants and sensitizes them to quinidine blockade. It is plausible that the Slack-Na<sub>v</sub>1.6 interaction contributes to the therapeutic role of quinidine in the treatment of KCNT1-related epilepsy.

## Viral expression of Slack's C-terminus prevents Slack<sup>G269S</sup>-induced seizures

Having established that blocking Na<sub>v</sub>1.6-mediated sodium influx significantly reduced Slack current amplitudes (Fig. 2D,G), we found that the heterozygous knockout of Na<sub>v</sub>1.6 significantly reduced the afterhyperpolarization amplitude in murine hippocampal neurons (Fig. S8), together indicating that Na<sub>v</sub>1.6 activates native Slack through providing Na<sup>+</sup>. We therefore assumed that disruption of the Slack-Na<sub>v</sub>1.6 interaction should reduce the amount of Na<sup>+</sup> in the close vicinity of epilepsy-related Slack mutant variants and thereby counter the increased current amplitudes of these Slack mutant variants. Pursuing this, we disrupted the

Slack- $\text{Na}_V$  interaction by overexpressing Slack's C-terminus (to compete with Slack) and measured whole-cell current densities. In HEK293 cells co-expressing epilepsy-related Slack mutant variants (G288S, R398Q)<sup>(41),(42)</sup> and  $\text{Na}_V$ , expression of Slack's C-terminus significantly reduced whole-cell current densities of Slack<sup>G269S</sup> and Slack<sup>R398Q</sup> (Fig. 7A,B), supporting that disrupting the Slack- $\text{Na}_V$ 1.6 interaction can indeed reduce current amplitudes of Slack mutant variants, which may protect against seizures induced by Slack mutant variants.

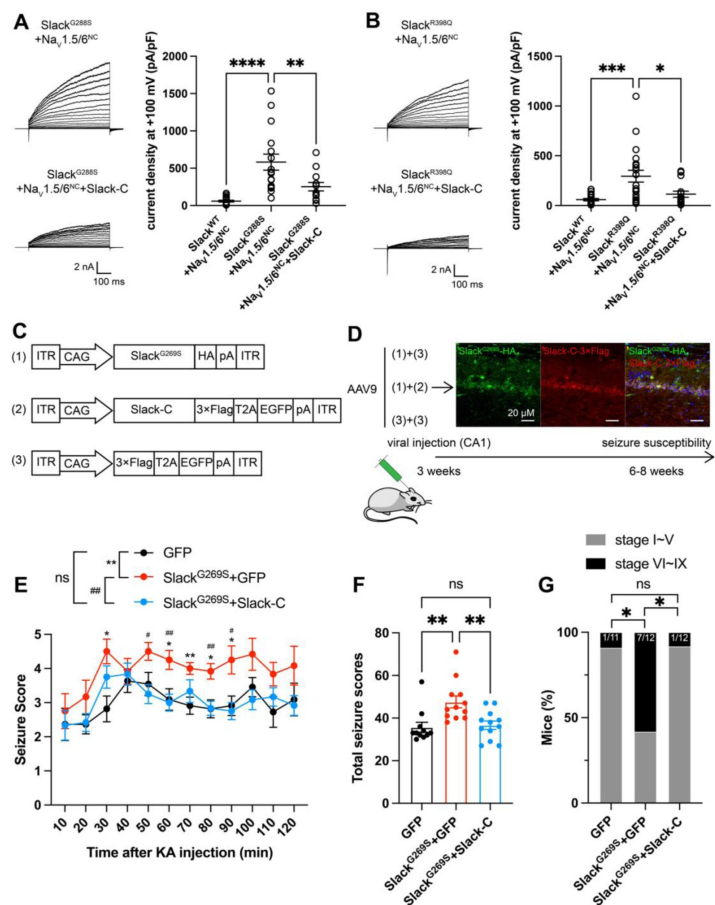


Figure 7.

## Viral expression of Slack's C-terminus prevents Slack<sup>G269S</sup>-induced seizures.

(A-B) The current densities of Slack mutant variants (Slack<sup>G288S</sup> [A] and Slack<sup>R398Q</sup> [B]) upon co-expression with  $\text{Na}_V$ 1.5/6<sup>NC</sup> in HEK293T cells were reduced by additional expression of Slack's C-terminus. Left, example current traces from HEK293T cells co-expressing Slack mutant variants and  $\text{Na}_V$ 1.5/6<sup>NC</sup> or co-expressing Slack mutant variants,  $\text{Na}_V$ 1.5/6<sup>NC</sup>, and Slack's C-terminus. Right, summarized current densities at +100 mV. \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ ; one-way ANOVA followed by Bonferroni's post hoc test. (C) Architecture for expression cassettes of AAVs. (D) Top, Immunofluorescence of HA-tagged Slack<sup>G269S</sup> (green), 3xFlag-tagged Slack's C-terminus (red), and DAPI (blue) at 5 weeks after viral injection of Slack<sup>G269S</sup> with Slack's C-terminus into CA1 of mice. Bottom, study design and timeline for the stereotactic injection model. (E) Time-course of KA-induced seizure stage changes at 10-min intervals based on a modified Racine, Pinal, and Rovner scale (please refer to Methods for further details). The number of mice used: "GFP" control group (n = 11), "Slack<sup>G269S</sup>+GFP" group (n = 12), "Slack<sup>G269S</sup>+Slack-C" group (n = 12). "GFP" vs.

"Slack<sup>G269S</sup>+GFP":  $F_{(1,21)} = 10.48$ ,  $p = 0.0040$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ ; "Slack<sup>G269S</sup>+GFP" vs. "Slack<sup>G269S</sup>+Slack-C":  $F_{(1,22)} = 10.30$ ,  $p = 0.0040$ , #  $p < 0.05$ , ##  $p < 0.01$ . "GFP" vs. "Slack<sup>G269S</sup>+Slack-C":  $F_{(1,21)} = 0.09574$ ,  $p = 0.7600$ . Repeated two-way ANOVA followed by Bonferroni's post hoc test. (F) Total seizure score per mouse over the 2 h after KA injection of these three groups. \*  $p < 0.05$ , \*\*  $p < 0.01$ ; one-way ANOVA followed by Bonferroni's post hoc test. (G) The percentage of mice with stage VI-IX seizures over the 2 h after KA injection in each group. \*  $p < 0.05$ ; Fisher's exact test.

We next induced an *in vivo* epilepsy model by introducing a Slack G269S variant into C57BL/6N mice using adeno-associated virus (AAV) injection to mimic the human Slack mutation G288S<sup>(41)</sup>. Specifically, we delivered stereotactic injections of AAV9 containing expression cassettes for Slack<sup>G269S</sup> (or GFP negative controls) into the hippocampal CA1 region of 3-week-old C57BL/6N mice (Fig. 7C,D). At 3-5 week intervals after AAV injection, we quantified the seizure susceptibility of mice upon the induction of a classic kainic acid (KA) model of temporal lobe epilepsy<sup>(43),(44)</sup>. In this model, seizures with stage IV or higher (as defined by a modified Racine, Pinal, and Rovner scale<sup>(45)</sup>) are induced in rodents by intraperitoneal administration of 28 mg/kg KA.



We assessed a time course of KA-induced seizure stages at 10-min intervals and found that viral expression of Slack<sup>G269S</sup> resulted in faster seizure progression in mice compared to control GFP expression (Fig. 7E). We calculated the total seizure score per mouse to assess seizure severity<sup>(46)</sup>. Slack<sup>G269S</sup>-expressing mice showed significantly higher seizure severity than GFP-expressing mice (Fig. 7F). The percentage of mice with stage VI~IX seizures also increased, from 9.1% in GFP-expressing control mice to 58.3% in the Slack<sup>G269S</sup>-expressing mice (Fig. 7G). These results support that viral expression of Slack<sup>G269S</sup> significantly increases seizure susceptibility in mice.

To evaluate the potential therapeutic effects of disrupting the Slack-Na<sub>v</sub>1.6 interaction, we delivered two AAV9s (one for Slack<sup>G269S</sup> and one for Slack's C-terminus [residues 326-1238]) into the CA1 region of mice (Fig. 7D). Viral expression of Slack's C-terminus in Slack<sup>G269S</sup>-expressing mice significantly decreased seizure progression, seizure severity, and the percentage of mice experiencing stage VI~IX seizures (Fig. 7E-G). These results support that viral expression of Slack's C-terminus can prevent Slack<sup>G269S</sup>-induced seizures in mice, thus showcasing that using Slack's C-terminus to disrupt the Slack-Na<sub>v</sub>1.6 interaction is a promising therapeutic intervention to treat KCNT1-related epilepsy.

## Discussion

We here found that Na<sub>v</sub>1.6's N- and C-termini bind to Slack's C-terminus and sensitize Slack to quinidine blockade via Na<sub>v</sub>1.6-mediated transient sodium currents. These results suggest that the pharmacological blocking effects of a channel blocker are not exclusively mediated by the channel *per.se.*, but modulated by channel's interacting proteins. Moreover, we show that viral expression of Slack's C-terminus can rescue the increased seizure susceptibility and confer protection against Slack<sup>G269S</sup>-induced seizures in mice.

At resting membrane potential, the intracellular sodium concentration ([Na<sup>+</sup>]<sub>in</sub>) in neurons (equal to ~10 mM) is too low to effectively activate Slack (K<sub>d</sub> of 66 mM)<sup>(1),(20)</sup>. Slack is functionally coupled to sodium influx, which is known to be mediated by ion channels and receptors, including Na<sub>v</sub>, AMPARs, and NMDARs<sup>(21)–(23),(47)</sup>. Such Na<sup>+</sup> sources can provide both the membrane depolarization and the Na<sup>+</sup> entry known to be required for Slack activation, enabling Slack to contribute both to action potential repolarization during neuronal high-frequency firing<sup>(7),(48)</sup> and to regulating excitatory postsynaptic potential (EPSP) at post-synaptic neurons<sup>(23)</sup>. Our results support that Slack and Na<sub>v</sub>1.6 form a channel complex, while also implying that Na<sub>v</sub>1.6-mediated sodium influx increases the Na<sup>+</sup> concentration in the close vicinity of Slack to activate Slack. As a low threshold Na<sub>v</sub> channel subtype, Na<sub>v</sub>1.6 has been reported to dominate the initiation and propagation of action potentials in axon initial segments (AIS) in excitatory neurons<sup>(36),(49)–(51)</sup>. The activation of Slack by Na<sub>v</sub>1.6 at AIS has multiple impacts, including ensuring the timing of the fast-activated component of Slack currents, regulating the action potential amplitude, and apparently contributing to intrinsic neuronal excitability<sup>(29)</sup>.

An interesting question arises from our observation that Slack's sensitivity to quinidine blockade is enhanced by I<sub>NaT</sub> but not I<sub>NaP</sub>: what can explain the distinct I<sub>NaT</sub> and I<sub>NaP</sub> contributions? We speculate that with physical modulation by Na<sub>v</sub>1.6, I<sub>NaT</sub> may elicit a specific open conformation of Slack that brings its quinidine binding pocket into a high affinity state, which could lead to a substantial increase in Slack's sensitivity to quinidine blockade. The possibility of this hypothetical open conformation is supported by previous reports of the presence of subconductance states detected in single-channel recordings of *Xenopus* oocytes expressing Slack channels<sup>(52)</sup>, which implies that there are multiple open

conformations of Slack. Although only one open conformation of Slack has been observed in cryo-EM, the gap between maximum conformational open probability ( $\sim 1.0$ ) and maximum functional open probability ( $\sim 0.7$ ) implies a subclass within this class of open channels<sup>(53)</sup>.

As previously mentioned, gain-of-function Slack mutant variants have been linked to a broad spectrum of epileptic disorders that are accompanied by intellectual disabilities and both psychomotor and developmental defects<sup>(18),(54)</sup>. Given that many patients are refractory or non-responsive to conventional anticonvulsants<sup>(12),(54),(55)</sup>, and considering the limited success of quinidine in clinical treatment of KCNT1-related epilepsy, inhibitors targeting Slack are needed urgently<sup>(18)</sup>. Several small-molecule inhibitors against Slack have been reported, providing informative starting points for drug development efforts with KCNT1-related epilepsy<sup>(28),(56),(57)</sup>. However, our discovery of the Slack- $\text{Na}_v1.6$  complex challenges the traditional view that Slack acts as an isolated target in KCNT1-related epilepsy<sup>(18)</sup>. Indeed, our study supports that co-expression of Slack and  $\text{Na}_v1.6$  in heterologous cell models should be performed when analyzing clinically relevant Slack mutations and when screening anti-epileptic drugs for use in treating KCNT1-related disorders.

Genotype-phenotype analysis has shown that ADSHE-related mutations are clustered in the regulator of conductance of  $\text{K}^+$  (RCK2) domain; while EIMFS-related mutations do not show a particular pattern of distribution<sup>(11)</sup>. All functionally tested Slack mutant variants show gain-of-function phenotypes, with increased Slack currents<sup>(9),(11),(18)</sup>. Further, the epilepsy-related Slack mutant variants confer their gain-of-function phenotypes through two molecular mechanisms: increasing maximal channel open probability ( $P_{\text{max}}$ ) or increasing sodium sensitivity ( $K_d$ ) of Slack<sup>(58)</sup>. Both  $P_{\text{max}}$  and  $K_d$  are highly sensitive to  $[\text{Na}^+]_{\text{in}}$ ; these are respectively analogous to the efficacy and potency of  $[\text{Na}^+]_{\text{in}}$  on Slack currents<sup>(58)</sup>. Notably, several Slack mutant variants show gain-of-function phenotypes only at high  $[\text{Na}^+]_{\text{in}}$  (e.g. 80 mM)<sup>(58)</sup>. These results indicate that the gain-of-function phenotype of epilepsy-related Slack mutant variants is aggravated by high  $[\text{Na}^+]_{\text{in}}$ . Our discovery of functional coupling between  $\text{Na}_v1.6$  and Slack presents a plausible basis for how  $\text{Na}_v1.6$ -mediated sodium influx can increase  $[\text{Na}^+]_{\text{in}}$  and thus apparently aggravate the gain-of-function phenotype of Slack mutant variants. Therefore, it makes sense that disruption of the Slack- $\text{Na}_v1.6$  interaction by overexpressing Slack's C-terminus reduces the current amplitudes of gain-of-function Slack mutant variants (Fig. 7A,B). Our successful demonstration that viral expression of Slack's C-terminus prevents epilepsy-related Slack<sup>G269S</sup>-induced seizures in mice (Fig. 7E-G) warrants further translational evaluation for developing therapeutic interventions to treat KCNT1-related epilepsy.

## Methods

### Animals

C57BL/6 mice were purchased from Charles River Laboratories.  $\text{Na}_v1.6$  knockout C3HeB/FeJ mice were generous gifts from Professor Yousheng Shu at Fudan University. All animals were housed on a 12-hour light/dark cycle with *ad libitum* access to food and water. All procedures related to animal care and treatment were approved by the Peking University Institutional Animal Care and Use Committee and met the guidelines of the National Institute of Health Guide for the Care and Use of Laboratory Animals. Each effort was made to minimize animal suffering and the number of animals used. The experiments were blind to viral treatment condition during behavioral testing.

## Antibodies and Reagents

Commercial antibodies used were: anti-AnkG (Santa Cruz), anti-Slack (NeuroMab), anti-Nav1.2 (Alomone), anti-Nav1.6 (Alomone), anti-HA (Abbkine), anti-Flag (Abbkine), anti- $\beta$ -Actin (Biodragon), HRP Goat Anti-Mouse IgG LCS (Abbkine), HRP Mouse Anti-Rabbit IgG LCS (Abbkine), Alexa Fluor 488-AffinityPure Fab Fragment Donkey anti-rabbit IgG (Jackson), Alexa Fluor 594 Donkey anti-mouse IgG (Yeaston). GPCR Extraction Reagent was from Pierce, NP40 lysis buffer was from Beyotime, protease inhibitor mixture cocktail was from Roche Applied Science, rabbit IgG and mouse IgG was from Santa Cruz, and Protein G Dynabeads were from Invitrogen. Tetrodotoxin was from Absin Bioscience. Quinidine was from Macklin, and riluzole was from Meilunbio. All other reagents were purchased from Sigma-Aldrich.

## Molecular cloning

Human Slack-B (Ref Seq: NM\_020822.3) and Nav1.6 (Ref Seq: NM\_014191.4) were subcloned into the modified pcDNA3.1(+) vector using Gibson assembly. All mutations and chimeras of ion channels were also constructed using Gibson assembly. For GST pull down assay, the segments of Slack and Nav1.6 were subcloned into pcDNA3.1(+) and pGEX-4T-1 vector, respectively. For FRET experiments, mVenus-tag was fused to the C-terminus of Nav1.6 sequence, and mTFP1-tag was also fused to the C-terminus of Slack sequence.

## Immunoprecipitation

The brain tissues or HEK293T cells co-expressing full-length or fragments of Slack and Nav1.6 were homogenized and lysed in GPCR Extraction Reagent (Pierce) with cocktail for 30 min at 4 °C. The homogenate was centrifuged for 20 min at 16 000 g and 4 °C to remove cell debris and then supernatant was incubated with 5  $\mu$ g Slack antibody (NeuroMab) or Nav1.6 antibody (Alomone) for 12 h at 4 °C with constant rotation. 40  $\mu$ l of protein G Dynabeads (Invitrogen) was then added and the incubation was continued until the next day. Beads were then washed three times with NP40 lysis. Between washes, the beads were collected by DynaMag. The remaining proteins were eluted from the beads by re-suspending the beads in 1 $\times$ SDS-PAGE loading buffer and incubating for 30 min at 37 °C. The resultant materials from immunoprecipitation or lysates were then subjected to western blot analysis.

## Western blot analysis

Proteins suspended in 1 $\times$ SDS-PAGE loading buffer were denatured for 30 min at 37 °C. Then proteins were loaded on 6% or 8% sodium dodecyl sulphate–polyacrylamide gel electrophoresis and transferred onto nitrocellulose filter membrane (PALL). Non-specific binding sites were blocked with Tris-buffered saline-Tween (0.02 M Tris, 0.137 M NaCl and 0.1% Tween 20) containing 5% non-fat dried milk. Subsequently, proteins of interest were probed with primary antibodies for overnight at 4 °C. After incubation with a secondary antibody, immunoreactive bands were visualized using HRP Substrate Peroxide Solution (Millipore) according to the manufacturer's recommendation.

## GST pull down assay

Plasmids encoding GST-fused Nav1.6 segments were transformed into BL21(DE3). After expressing recombinant proteins induced by overnight application of isopropyl-1-thio- $\beta$ -D-galactopyranoside (IPTG) (0.1mM) at 25 °C, the bacteria were collected, lysed and incubated

with GSH beads using BeaverBeads GSH kit (Beaver) according to the manufacturer's instruction.

Plasmids encoding KCNT1 channel or HA-tagged KCNT1 segments were transfected into HEK293T cells using lipofectamine 2000 (Introvigen). 40 h after transfection, cells were lysed in NP40 lysis buffer with inhibitor cocktail, and then centrifugated at 4 °C, 15000 g for 20 min. The supernatants were incubated with protein-bound beads. The protein-bound beads were washed by washing buffer (50 mM Tris-HCl pH = 7.4, 120 mM NaCl, 2 mM EGTA, 2 mM DTT, 0.1% triton X-100, and 0.2% Tween-20) for 5 min 3 times and then denatured with 1×SDS-PAGE loading buffer and incubating for 30 min at 37 °C. The resultant materials were subjected to western blot analysis.

## Cell culture

The human embryonic kidney cells (HEK293 and HEK293T) were maintained in Dulbecco's Modified Eagle Medium (DMEM, Gibco) supplemented with 15% Fetal Bovine Serum (FBS, PAN-Biotech) at 37°C and 5% CO<sub>2</sub>. Primary cortical neurons were prepared from either sex of postnatal homozygous Na<sub>v</sub>1.6 knockout C3HeB/FeJ mice and the wild-type littermate controls. After the mice were decapitated, the cortices were removed and separated from the meninges and surrounding tissue. Tissues were digested in 2 mg/mL Papain (Aladdin) containing 2 µg/mL DNase I (Psaotong) for 30 min followed by centrifugation and resuspension. Subsequently, the cells were plated on poly-D-lysine (0.05 mg/mL) pre-coated glass coverslips in plating medium (DMEM containing 15% FBS), at a density of 4×10<sup>5</sup> cells/mL, and cultivated at 37°C and 5% CO<sub>2</sub> in a humidified incubator. 5 h after plating, the medium was replaced with Neurobasal Plus medium (Invitrogen) containing 2% v/v B-27 supplement (Invitrogen), 2 mM Glutamax (Invitrogen), 50 U/mL penicillin and streptomycin (Life Technologies). The primary neurons were grown 6-10 days before electrophysiological recordings with half of the media replaced every three days.

## Voltage-clamp recordings

The plasmids expressing full-length or fragments of Slack and Na<sub>v</sub> channels (excluding full-length Na<sub>v</sub>1.6) were co-transfected into HEK293 cells using lipofectamine 2000 (Introvigen). To co-express Slack with Na<sub>v</sub>1.6, the plasmid expressing Slack was transfected into a stable HEK293 cell line expressing Na<sub>v</sub>1.6. 18-36 h after transfection, voltage-clamp recordings were obtained using a HEKA EPC-10 patch-clamp amplifier (HEKA Electronic) and PatchMaster software (HEKA Electronic). For all whole-cell patch clamp experiments in HEK cells except for data presented in Fig. S2d, the extracellular recording solution contained (in mM): 140 NaCl, 3 KCl, 1 CaCl<sub>2</sub>, 1 MgCl<sub>2</sub>, 5 glucose, 10 HEPES, 1 Tetraethylammonium chloride (310 mOsm/L, pH 7.30 with NaOH). The recording pipette intracellular solution (5 mM Na) contained (in mM): 100 K-gluconate, 30 KCl, 15 Choline-Cl, 5 NaCl, 10 glucose, 5 EGTA, 10 HEPES (300 mOsm/L, pH 7.30 with KOH). For data presented in Fig. S2d, the extracellular recording solution remains the same as above, the pipette intracellular solution contained (in mM): 140 CsF, 10 NaCl, 5 EGTA, 10 HEPES (pH 7.30 with NaOH), indicating that the unusual right shifts (15~20 mV) in voltage dependence of Na<sub>v</sub>1.6 were induced by components in pipette solution, not recording system errors. For primary cortical neurons, intracellular solution (0 mM Na) was used to prevent the activation of sodium-activated potassium channels by basal intracellular sodium ions. NaCl in the intracellular solution was replaced with choline chloride in an equimolar concentration. For inside out patch-clamps, the bath solution contained (in mM): 140 NaCl, 1 EDTA, 10 HEPES and 2 MgCl<sub>2</sub> (310 mOsm/L, pH 7.30 with NaOH). Pipette solution contained (in mM): 130 KCl, 1 EDTA, 10 HEPES and 2 MgCl<sub>2</sub> (300 mOsm/L, pH 7.30 with KOH). The pipettes were fabricated by a DMZ Universal Electrode puller (Zeitz Instruments) using borosilicate glass, with a resistance of 1.5-3.5 MΩ



for whole-cell patch clamp recordings and 8.0-10.0 MΩ for inside-out patch clamp recordings. All experiments were performed at room temperature. The concentration-response curves were fitted to four-parameter Hill equation:

$$Y = Bottom + (Top - Bottom)/(1 + 10^{X-lgIC_{50}}) \quad (1)$$

where Y is the value of  $I_{Quinidine}/I_{Control}$ , Top is the maximum response, Bottom is the minimum response, X is the lg of concentration, and  $IC_{50}$  is the drug concentration producing the half-maximum response. Significance of fitted  $IC_{50}$  values compared to control was analyzed using extra sum-of-squares F test.

For data presented in fig. S3, cells were excluded from analysis if series resistance > 5 MΩ and series resistance compensation was set to 70%~90%.

The time constants ( $\tau$ ) of activation were fitted with a single exponential equation:

$$I(t) = Offset + A * \exp(-\frac{t}{\tau}) \quad (2)$$

where I is the current amplitude, t is time, offset represents the asymptote of the fit, and A represents the amplitude for the activation or inactivation.

Steady-state fast inactivation (I-V) and conductance-voltage (G-V) relationships were fitting with Boltzmann equations:

$$I/I_{max} = 1/(1 + \exp(\frac{V_m - V_{1/2}}{k})) \quad (3)$$

$$G/G_{max} = 1/(1 + \exp(\frac{V_m - V_{1/2}}{k})) \quad (4)$$

$$G = I/(V_m - E_{Na}) \quad (5)$$

where I is the peak current, G is conductance,  $V_m$  is the stimulus potential,  $V_{1/2}$  is the midpoint voltage,  $E_{Na}$  is the equilibrium potential, and k is the slope factor. Significance of fitted  $V_{1/2}$  compared to control was analyzed using extra sum-of-squares F test.

Recovery from fast inactivation data were fitted with a single exponential equation:

$$I/I_{max} = A * (1 - \exp(-\frac{t}{\tau})) \quad (6)$$

Where I is the peak current of test pulse,  $I_{max}$  is the peak current of first pulse, A is the proportional coefficient, t is the delay time between the two pulses, and  $\tau$  is the time constant of recovery from fast inactivation.

## Acute slice preparation and current-clamp recordings

Horizontal slices containing hippocampus were obtained from 6-8 weeks old heterozygous  $Na_v1.6$  knockout ( $Scn8a^{+/-}$ ) C3HeB/FeJ mice and the wild-type littermate controls. In brief, animals were anesthetized and perfused intracardially with ice-cold modified “cutting solution” containing (in mM): 110 choline chloride, 2.5 KCl, 0.5  $CaCl_2$ , 7  $MgCl_2$ , 25  $NaHCO_3$ , 1.25  $NaH_2PO_4$ , 10 glucose; bubbled continuously with 95% $O_2$ /5% $CO_2$  to maintain PH at 7.2.

The brain was then removed and submerged in ice-cold “cutting solution”. Next, the brain was cut into 300  $\mu\text{m}$  slices with a vibratome (WPI). Slices were incubated in oxygenated (95%  $\text{O}_2$  and 5%  $\text{CO}_2$ ) “recording solution” containing (in mM): 125 NaCl, 2.5 KCl, 2  $\text{CaCl}_2$ , 2  $\text{MgCl}_2$ , 25  $\text{NaHCO}_3$ , 1.25  $\text{NaH}_2\text{PO}_4$ , 10 glucose (315 mOsm/L, pH 7.4, 37°C) for 30 min, and stored at room temperature.

Slices were subsequently transferred to a submerged chamber containing “recording solution” maintained at 34–36 °C. Whole-cell recordings were obtained from hippocampal CA1 neurons under a  $\times 60$  water-immersion objective of an Olympus BX51WI microscope (Olympus). Pipettes had resistances of 5–8 M $\Omega$ . For current-clamp recordings, the external solution (unless otherwise noted) was supplemented with 0.05 mM (2R)-amino-5-phosphonovaleric acid (APV), 0.01 mM 6-cyano-7-nitro-quinoxaline-2,3-dione, 0.01 mM bicuculline and 0.001 mM CGP 55845, and internal pipette solution containing (in mM): 118 KMeSO<sub>4</sub>, 15 KCl, 10 HEPES, 2  $\text{MgCl}_2$ , 0.2 EGTA, and 4 Na<sub>2</sub>ATP, 0.3 Tris-GTP, 14 Tris-phosphocreatinin (pH 7.3 with KOH).

In our whole-cell current-clamp recordings with an Axon 700B amplifier (Molecular Devices), we initially applied a 100 ms, 20 pA test pulse to the recording neurons right after breaking into the whole-cell configuration. A fast-rising component and a slow-rising component of voltage response were clearly visible. Then we zoomed into the fast-rising component of the voltage responses and turned up the pipette capacitance neutralization slowly to shorten the rise time of the fast-rising component until the oscillations of voltage responses appeared. Subsequently, we decreased the capacitance compensation just until the oscillations disappeared. For bridge balance of current-clamp recordings, we increased the value of bridge balance slowly until the fast component of the voltage response disappeared, and the slow-rising component appeared to rise directly from the baseline. Series resistance and pipette capacitance were compensated using the bridge balance and pipette capacitance neutralization options in the Multiclamp 700B command software (Molecular Devices). The bridge balance value was between 20 and 30 M $\Omega$  and the pipette capacitance neutralization value was between 3 and 5 pF. For in vitro experiments, the cells were selected by criteria based on hippocampal CA1 cell morphology and electrophysiological properties in the slices. Electrophysiological recordings were made using a Multiclamp 700B amplifier (Molecular Devices). Recordings were filtered at 10 kHz and sampled at 50 kHz. Data were acquired and analyzed using pClamp10.0 (Molecular Devices). Series resistance was in the order of 10–30 M $\Omega$  and was approximately 60–80% compensated. Recordings were discarded if the series resistance increased by more than 20% during the time course of the recordings.

## Fluorescence Imaging and FRET Quantification

The spectroscopic imaging was built upon a Nikon TE2000-U microscope. The excitation light was generated by an Ar laser. The fluorescent protein mVenus fused to Na<sub>v</sub>1.6 and mTFP1 fused to Slack were excited by laser line at 500 and 400–440 nm, respectively. The duration of light exposure was controlled by a computer-driven mechanical shutter (Uniblitz). A spectrograph (Acton SpectraPro 2150i) was used in conjunction with a charge coupled device (CCD) camera (Roper Cascade 128B). In this recording mode two filter cubes (Chroma) were used to collect spectroscopic images from each cell (excitation, dichroic): cube I, D436/20, 455dclp; cube II, HQ500/20, Q515lp. No emission filter was used in these cubes. Under the experimental conditions, auto fluorescence from untransfected cells was negligible. Fluorescence imaging and analysis were done using the MetaMorph software (Universal Imaging). User-designed macros were used for automatic collection of the bright field cell image, the fluorescence cell image, and the spectroscopic image. Emission spectra were collected from the plasma membrane of the cell by positioning the spectrograph slit across a cell and recording the fluorescence intensity at the position corresponding to the membrane

region (Fig. 2e, dotted lines in red); the same slit position applied to both the spectrum taken with the mTFP1 excitation and the spectrum taken with the mVenus excitation. Using this approach, the spectral and positional information are well preserved, thus allowing reliable quantification of FRET efficiency specifically from the cell membrane. Spectra were corrected for background light, which was estimated from the blank region of the same image.

FRET data was quantified in two ways. First, the FRET ratio was calculated from the increase in mVenus emission due to energy transfer as described in the previous study.<sup>(59)</sup> Briefly, mTFP1 emission was separated from mVenus emission by fitting of standard spectra acquired from cells expressing only mVenus or mTFP1. The fraction of mVenus-tagged molecules that are associated with mTFP1-tagged molecules,  $Ab$ , is calculated as

$$Ab = 1/(1 + K_D/[D_{free}]) \quad (7)$$

where  $K_D$  is the dissociation constant and  $[D_{free}]$  is the concentration of free donor molecules. Note that

$$FRET \text{ Ratio} = 1 + Ab * (FRET \text{ Ratio}_{max} - 1) \quad (8)$$

Regression analysis was used to estimate  $Ab$  in individual cells. From each cell, the FRET ratio<sub>exp</sub> was experimentally determined. The predicted  $Ab$  value was then computed by adjusting two parameters, FRET Ratio<sub>max</sub> and apparent  $K_D$ .  $Ab$  was in turn used to give a predicted FRET ratio<sub>predicted</sub>. By minimizing the squared errors  $(FRET \text{ ratio}_{exp} - FRET \text{ ratio}_{predicted})^2$ ,  $K_D$  was determined.

Second, apparent FRET efficiency was also calculated from the enhancement of mVenus fluorescence emission due to energy transfer<sup>(59)–(62)</sup> using a method as previously described.<sup>(63)</sup> Briefly, Ratio  $A_0$  and Ratio  $A$  were measured to calculate FRET efficiency. Ratio  $A_0$  represents the ratio between tetramethylrhodamine maleimide emission intensities (in the absence of fluorescein maleimide) upon excitation at the donor and acceptor excitation wavelengths<sup>(62)–(64)</sup>, and was calculated in the present study at the mVenus peak emission wavelength. A particular advantage of quantifying Ratio  $A_0$  for FRET measurement is that changes in fluorescence intensity caused by many experimental factors can be cancelled out by the ratiometric measurement. A similar ratio, termed Ratio  $A$ , was determined in the presence of mTFP1 in the same way as Ratio  $A_0$ . If FRET occurred, the Ratio  $A$  value should be higher than Ratio  $A_0$ ; the difference between Ratio  $A$  and Ratio  $A_0$  was directly proportional to the FRET efficiency by the factor of extinction coefficient ratio of mTFP1 and mVenus.<sup>(62)–(64)</sup>

## Immunostaining

After deep anesthesia with sodium pentobarbital, mice were sacrificed by perfusion with 0.5% paraformaldehyde and 0.5% sucrose (wt/vol) in 0.1 M phosphate buffer (pH 7.4). The brain was removed and post-fixed in the same fixative for 2 h, and subsequently immersed in 30% sucrose in 0.1 M phosphate buffer for 48 h. Cryostat coronal sections (20  $\mu$ m) were obtained using a freezing microtome (Leica). The sections were rinsed in 0.01 M phosphate-buffered saline (PBS, pH 7.4), permeabilized in 0.5% Triton X-100 in PBS for 30 min, and incubated in a blocking solution (5% BSA, 0.1% Triton X-100 in PBS, vol/vol) at 20–25 °C for 2 h, followed by overnight incubation at 4 °C with primary antibody to AnkG (1:100, Santa Cruz, sc-31778), Slack (1:100, NeuroMab, 73-051), Na<sub>v</sub>1.2 (1:200, Alomone, ASC-002), Na<sub>v</sub>1.6 (1:200, Alomone, ASC-006), Flag (1:500, Abbkine, ABT2010), and HA (1:500, Abbkine,

ABT2040) in blocking solution. After a complete wash in PBS, the sections were incubated in Alexa 488-conjugated donkey anti-rabbit IgG and Alexa 594 donkey anti-mouse IgG in blocking solution at 20–25 °C for 2 h. The sections were subsequently washed and rinsed in Dapi solution. Images were taken in the linear range of the photomultiplier with a laser scanning confocal microscope (ZEISS LSM 510 META NLO).

## Adeno-associated virus construction and injection

The adeno-associated viruses (AAVs) and the negative GFP control were from Shanghai GeneChem. Co., Ltd. The full-length Slack<sub>G269S</sub> sequence (1-1238aa) was ligated into modified CV232 (CAG-MCS-HA-Poly A) adeno-associated viral vector. The Slack's C-terminus sequence (326-1238aa) and the negative control were ligated into GV634 (CAG-MCS-3×Flag-T2A-EGFP-SV40-Poly A) adeno-associated viral vector. The viruses (>10<sup>11</sup> TU/ml) were used in the present study.

For dorsal CA1 viral injection, C57BL/6J mice aged 3 weeks were anesthetized with isoflurane and placed in a stereotaxic apparatus (RWD Life Science Co., Ltd.). Using a 5 µL micro syringe (Hamilton) with a 30 gauge needle (RWD Life Science Co., Ltd.), 600 nL of the viruses was delivered at 10 nL/min by a micro-syringe pump (RWD Life Science Co., Ltd.) at the following site in each of the bilateral CA1 regions, using the stereotaxic coordinates: 2.5 mm (anterior-posterior) from bregma, 2 mm (medio-lateral), ± 1.5 mm (dorsal-ventral)<sup>(65)</sup>. The syringe was left in place for 5 min after each injection and withdrawn slowly. The exposed skin was closed by surgical sutures and returned to home cage for recovery. All the experiments were conducted after at least 3 weeks of recovery. All the mice were sacrificed after experiments to confirm the injection sites and the viral trans-infection effects by checking EGFP under a fluorescence microscope (ZEISS LSM 510 META NLO).

## Kainic acid-induced status epilepticus

KA (Sigma-Aldrich) was intraperitoneally administered to produce seizures with stage IV or higher. The dose of kainic acid used was 28 mg/kg for mice (6–8 weeks)<sup>(66),(67)</sup>. To assess epilepsy susceptibility, seizures were rated using a modified Racine, Pinal, and Rovner scale<sup>(45),(68)</sup>: (1) Facial movements; (2) head nodding; (3) forelimb clonus; (4) dorsal extension (rearing); (5) Loss of balance and falling; (6) Repeated rearing and falling; (7) Violent jumping and running; (8) Stage 7 with periods of tonus; (9) Dead. Seizures was terminated 2 h after onset with the use of sodium pentobarbital (30 mg/kg; Sigma-Aldrich).

## Statistical analysis

For in vitro experiments, the cells were evenly suspended and then randomly distributed in each well tested. For in vivo experiments, the animals were distributed into various treatment groups randomly. Statistical analyses were performed using GraphPad Prism 9 (GraphPad Software) and SPSS 26.0 software (SPSS Inc.). Before statistical analysis, variation within each group of data and the assumptions of the tests were checked. Comparisons between two independent groups were made using unpaired Student's two-tailed t test. Comparisons among nonlinear fitted values were made using extra sum-of-squares F test. Comparisons among three or more groups were made using one- or two-way analysis of variance followed by Bonferroni's post hoc test. No statistical methods were used to predetermine sample sizes but our sample sizes are similar to those reported previously in the field<sup>(37),(69)</sup>. All experiments and analysis of data were performed in a blinded manner by investigators who were unaware of the genotype or manipulation. \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ , \*\*\*\*  $p < 0.0001$ . All data are presented as mean ± SEM.

## Author contributions

T.Y. and Y.J. performed and analyzed voltage-clamp recordings. Y.W. performed and analyzed Western blotting, immunoprecipitation. S.X., C.P., and G.D. performed and analyzed pull down assay. Q.C. performed the immunostaining. H.Z. and F.Y. performed and analyzed the FRET imaging. H.S., N.L., and X.M. performed and analyzed the current clamp recordings. T.Y., H.Y., Z.G., and J.D. performed the molecular cloning. Z.H., T.Y., and Y.W. designed the experiments. Z.H., T.Y., and Y.J. wrote the manuscript. Z.H., F.Y., Y.Y., and Q.S. reviewed the manuscript.

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## Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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## Author information

### Tian Yuan

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

### Yifan Wang

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

### Yuchen Jin

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

### Shuai Xu

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Heng Zhang**

NHC and CAMS Key Laboratory of Medical Neurobiology, MOE Frontier Science Center for Brain Research and Brain-Machine Integration, School of Brain Science and Brain Medicine, Zhejiang University, Hangzhou, Zhejiang, 310058, China

**Qian Chen**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Na Li**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Xinyue Ma**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Huifang Song**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Chao Peng**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Hui Yang**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Ze Geng**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Jie Dong**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Guifang Duan**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Qi Sun**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China

**Yang Yang**

Department of Medicinal Chemistry and Molecular Pharmacology, College of Pharmacy, Purdue University, West Lafayette, IN 47907, USA  
ORCID iD: [0000-0001-6417-3654](https://orcid.org/0000-0001-6417-3654)

**Fan Yang**

Department of Biophysics, Kidney Disease Center of the First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, Zhejiang, 310058, China, NHC and CAMS Key Laboratory of Medical Neurobiology, MOE Frontier Science Center for Brain Research and Brain-Machine Integration, School of Brain Science and Brain Medicine, Zhejiang University, Hangzhou, Zhejiang, 310058, China

**Zhuo Huang**

State Key Laboratory of Natural and Biomimetic Drugs, Department of Molecular and Cellular Pharmacology, School of Pharmaceutical Sciences, Peking University Health Science Center, Beijing, 100191, China, IDG/McGovern Institute for Brain Research, Peking University, Beijing, 100871, China

**For correspondence:** [huangz@hsc.pku.edu.cn](mailto:huangz@hsc.pku.edu.cn)

ORCID iD: [0000-0001-9198-4778](https://orcid.org/0000-0001-9198-4778)

**Editors**

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**Stephan Pless**

University of Copenhagen, Denmark

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National Institute of Neurological Disorders and Stroke, National Institutes of Health, United States of America

**Reviewer #1 (Public Review):**

Despite numerous studies on quinidine therapies for epilepsies associated with GOF mutant variants of Slack, there is no consensus on its utility due to contradictory results. In this study Yuan *et al.* investigated the role of different sodium selective ion channels on the sensitization of Slack to quinidine block. The study employed electrophysiological approaches, FRET studies, genetically modified proteins and biochemistry to demonstrate that Nav1.6 N- and C-tail interacts with Slack's C-terminus and significantly increases Slack sensitivity to quinidine blockade *in vitro* and *in vivo*. This finding inspired the authors to investigate whether they could rescue Slack GOF mutant variants by simply disrupting the interaction between Slack and Nav1.6. They find that the isolated C-terminus of Slack can reduce the current amplitude of Slack GOF mutant variants co-expressed with Nav1.6 in HEK cells and prevent Slack induced seizures in mouse models of epilepsy. This study adds

to the growing list of channels that are modulated by protein-protein interactions, and is of great value for future therapeutic strategies.

I have a few comments with regard to how Nav1.6 sensitize Slack to block by quinidine.

It is not clear to me if the Slack induced current amplitude varies depending on the specific Nav subtype. To this end, it would be valuable to test if Slack open probability is affected by the presence of specific Nav subtypes. Nav induced differences in Slack current amplitude and open probability could explain why individual Nav subtypes show varied ability to sensitize Slack to quinidine blockade.

It has previously been shown that INaP (persistent sodium current) is important for inducing Slack currents. Here the authors show that INaT (transient sodium current) of Nav1.6 is necessary for the sensitization of Slack to quinidine block whereas INaP surprisingly has no effect. The authors then show that the N-tail together with C-tail of Nav1.6 can induce same effect on Slack as full-length Nav1.6 in presence of high intracellular concentrations of sodium. However, it is not clear to me how the isolated N- and C-tail of Nav1.6 can induce sensitization of Slack to quinidine by interacting with C-terminus of Slack, while sensitization also is dependent on INaT. The authors speculate on different slack open conformation, but one could speculate if there is a missing link, such as an un-identified additional interacting protein that causes the coupling.

## Reviewer #2 (Public Review):

This is a very interesting paper about the coupling of Slack and Nav1.6 and the insight this brings to the effects of quinidine to treat some epilepsy syndromes.

Slack is a sodium-activated potassium channel that is important to hyperpolarization of neurons after an action potential. Slack is encoded by KNCT1 which has mutations in some epilepsy syndromes. These types of epilepsy are treated with quinidine but this is an atypical antiseizure drug, not used for other types of epilepsy. For sufficient sodium to activate Slack, Slack needs to be close to a channel that allows robust sodium entry, like Nav channels or AMPA receptors. but more mechanistic information is not available. Of particular interest to the authors is what allows quinidine to be effective in reducing Slack.

In the manuscript, the authors show that Nav, not AMPA receptors, are responsible for Slack's sensitization to quinidine blockade, at least in cultured neurons (HeK293, primary cortical neurons). Most of the paper focuses on the evidence that Nav1.6 promotes Slack sensitivity to quinidine.

The paper is very well written although there are reservations about the use of non-neuronal cells or cultured primary neurons rather than a more intact system. I also have questions about the figures. Finally, riluzole is not a selective drug, so the limitations of this drug should be discussed. On a minor point, the authors use the term *in vivo* but there are no *in vivo* experiments.

## Reviewer #3 (Public Review):

Yuan et al., set out to examine the role of functional and structural interaction between Slack and Navs on the Slack sensitivity to quinidine. Through pharmacological and genetic means they identify Nav1.6 as the privileged Nav isoform in sensitizing Slack to quinidine. Through biochemical assays, they then determine that the C-terminus of Slack physically interacts with the N- and C-termini of Nav1.6. Using the information gleaned from the *in vitro*

experiments the authors then show that virally-mediated transduction of Slack's C-terminus lessens the extent of SlackG269S-induced seizures. These data uncover a previously unrecognized interaction between a sodium and a potassium channel, which contributes to the latter's sensitivity to quinidine.

The conclusions of this paper are mostly well supported by data, but some aspects of functional and structural studies in vivo as well as physically interaction need to be clarified and extended.

1. Immunolabeling of the hippocampus CA1 suggests sodium channels as well as Slac colocalization with AnkG (Fig 3A). Proximity ligation assay for Nav1.6 and Slac or a super-resolution microscopy approach would be needed to increase confidence in the presented colocalization results. Furthermore, coimmunoprecipitation studies on the membrane fraction would bolster the functional relevance of Nav1.6-Slac interaction on the cell surface.
2. Although hippocampal slices from Scn8a<sup>+/-</sup> were used for studies in Fig. S8, it is not clear whether Scn8a<sup>-/-</sup> or Scn8a<sup>+/-</sup> tissue was used in other studies (Fig 1J & 1K). It will be important to clarify whether genetic manipulation of Nav1.6 expression (Fig. 1K) has an impact on sodium-activated potassium current, level of surface Slac expression, or that of Nav1.6 near Slac.
3. Did the epilepsy-related Slac mutations have an impact on Nav1.6-mediated sodium current?
4. Showing the impact of quinidine on persistent sodium current in neurons and on Nav1.6-expressing cells would further increase confidence in the role of persistent sodium current on sensitivity of Slac to quinidine.

## Author Response:

### Reviewer #1 (Public Review):

*Despite numerous studies on quinidine therapies for epilepsies associated with GOF mutant variants of Slack, there is no consensus on its utility due to contradictory results. In this study Yuan et al. investigated the role of different sodium selective ion channels on the sensitization of Slack to quinidine block. The study employed electrophysiological approaches, FRET studies, genetically modified proteins and biochemistry to demonstrate that Nav1.6 N- and C-tail interacts with Slack's C-terminus and significantly increases Slack sensitivity to quinidine blockade in vitro and in vivo. This finding inspired the authors to investigate whether they could rescue Slack GOF mutant variants by simply disrupting the interaction between Slack and Nav1.6. They find that the isolated C-terminus of Slack can reduce the current amplitude of Slack GOF mutant variants co-expressed with Nav1.6 in HEK cells and prevent Slack induced seizures in mouse models of epilepsy. This study adds to the growing list of channels that are modulated by protein-protein interactions, and is of great value for future therapeutic strategies.*

*I have a few comments with regard to how Nav1.6 sensitize Slack to block by quinidine.*

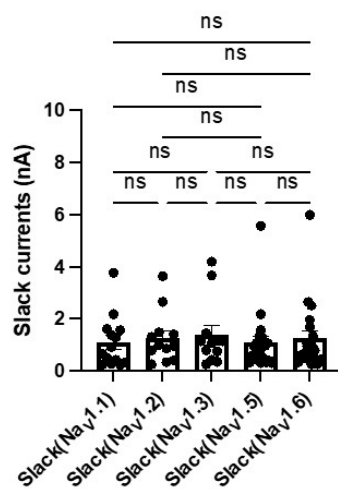


(1) It is not clear to me if the Slack induced current amplitude varies depending on the specific Nav subtype. To this end, it would be valuable to test if Slack open probability is affected by the presence of specific Nav subtypes. Nav induced differences in Slack current amplitude and open probability could explain why individual Nav subtypes show varied ability to sensitize Slack to quinidine blockade.

We appreciate the reviewer for raising this point. In order to address whether the whole-cell current amplitudes of Slack varies depending on the specific NaV subtype, we examined Slack current amplitudes upon co-expression of Slack with specific NaV subtypes in HEK293 cells. The results have shown that there are no significant differences in Slack current amplitudes upon co-expression of Slack with different NaV channel subtypes (Author response image 1), suggesting whole-cell Slack current amplitudes cannot explain the varied ability of NaV subtypes to sensitize Slack to quinidine blockade. To investigate the effect of different NaV channel subtypes on Slack open probability, we will perform the single-channel recordings in the future studies.

#### Author response image 1.

The amplitudes of Slack currents upon co-expression of Slack with specific NaV subtypes in HEK293 cells. ns,  $p > 0.05$ , one-way ANOVA followed by Bonferroni's post hoc test.



(2) It has previously been shown that INaP (persistent sodium current) is important for inducing Slack currents. Here the authors show that INaT (transient sodium current) of Nav1.6 is necessary for the sensitization of Slack to quinidine block whereas INaP surprisingly has no effect. The authors then show that the N-tail together with C-tail of Nav1.6 can induce same effect on Slack as full-length Nav1.6 in presence of high intracellular concentrations of sodium. However, it is not clear to me how the isolated N- and C-tail of Nav1.6 can induce sensitization of Slack to quinidine by interacting with C-terminus of Slack, while sensitization also is dependant on INaT. The authors speculate on different slack open conformation, but one could speculate if there is a missing link, such as an un-identified additional interacting protein that causes the coupling.

We fully agree the importance of investigating the detailed mechanism underlying the sensitization of Slack to quinidine blockade mediated by the N- and C-termini of Nav1.6. Regarding the possibility of additional interacting proteins ("missing link") that mediate the coupling between Slack and Nav1.6, our GST-pull down assays involving Slack and the N- and C-termini of Nav1.6 (Fig. S7) suggest a direct interaction between Slack and Nav1.6 channels. This finding leads us to consider the possibility of additional interacting proteins might be

excluded. In order to further address these questions, we plan to employ structural biological methods, such as cryo-electron microscopy (cryo-EM).

### **Reviewer #2 (Public Review):**

*This is a very interesting paper about the coupling of Slack and Nav1.6 and the insight this brings to the effects of quinidine to treat some epilepsy syndromes.*

*Slack is a sodium-activated potassium channel that is important to hyperpolarization of neurons after an action potential. Slack is encoded by KNCT1 which has mutations in some epilepsy syndromes. These types of epilepsy are treated with quinidine but this is an atypical antiseizure drug, not used for other types of epilepsy. For sufficient sodium to activate Slack, Slack needs to be close to a channel that allows robust sodium entry, like Na channels or AMPA receptors. but more mechanistic information is not available. Of particular interest to the authors is what allows quinidine to be effective in reducing Slack.*

*In the manuscript, the authors show that Nav, not AMPA receptors are responsible for Slack activation, at least in cultured neurons (HeK293, primary cortical neurons). Most of the paper focuses on the evidence that Nav1.6 promotes Slack sensitivity to quinidine.*

*(1) The paper is very well written although there are reservations about the use of non-neuronal cells or cultured primary neurons rather than a more intact system.*

We appreciate the reviewer's positive evaluation of our work. We acknowledge that utilizing a more intact system would provide valuable insights into the inhibitory effect of quinidine on Slack-Nav1.6. However, there are certain challenges associated with studying Slack currents in their entirety.

First, in our experiments, isolating Slack currents from Na<sup>+</sup>-activated K<sup>+</sup> currents in an intact system is challenging as selective inhibitors for Slick are currently unavailable. To address this, we propose using Slick gene knockout mice to specifically measure Slack currents under physiological conditions in the future investigations. Second, we have observed that the interaction between Slack and Nav1.6 primarily occurs at the axon initial segment of neurons. This poses a difficulty when using brain slices for measurements, as employing the whole-cell voltage-clamp technique to assess Slack at the axon initial segment may introduce systemic errors.

We believe that testing the pharmacological effects of quinidine on Slack-Nav1.6 in primary neurons remains the optimal approach. Although non-neuronal cells or cultured primary neurons may not fully replicate the complexity of an intact system, they still provide valuable insights into the interactions between Slack and Nav1.6, and the effects of quinidine.

*(2) I also have questions about the figures.*

We will make the necessary modifications and clarifications based on the reviewer's comments:

*(3) Finally, riluzole is not a selective drug, so the limitations of this drug should be discussed.*

We thank the reviewer for raising this point. We will discuss the limitations of riluzole in our revised version of the manuscript.

*(4) On a minor point, the authors use the term in vivo but there are no in vivo experiments.*

We thanks the reviewer for raising this point. In our experiments, although we did not conduct experiments directly in living organisms, our results demonstrated the co-immunoprecipitation of Nav1.6 with Slack in homogenates from mouse cortical and

hippocampal tissues (Fig. 3C). This result may support that the interaction between Slack and NaV1.6 occurs *in vivo*.

### Reviewer #3 (Public Review):

*Yuan et al., set out to examine the role of functional and structural interaction between Slack and NaVs on the Slack sensitivity to quinidine. Through pharmacological and genetic means they identify NaV1.6 as the privileged NaV isoform in sensitizing Slack to quinidine. Through biochemical assays, they then determine that the C-terminus of Slack physically interacts with the N- and C-termini of NaV1.6. Using the information gleaned from the *in vitro* experiments the authors then show that virally-mediated transduction of Slack's C-terminus lessens the extent of SlackG269S-induced seizures. These data uncover a previously unrecognized interaction between a sodium and a potassium channel, which contributes to the latter's sensitivity to quinidine.*

*The conclusions of this paper are mostly well supported by data, but some aspects of functional and structural studies *in vivo* as well as physically interaction need to be clarified and extended.*

*(1) Immunolabeling of the hippocampus CA1 suggests sodium channels as well as Slack colocalization with AnkG (Fig 3A). Proximity ligation assay for NaV1.6 and Slack or a super-resolution microscopy approach would be needed to increase confidence in the presented colocalization results. Furthermore, coimmunoprecipitation studies on the membrane fraction would bolster the functional relevance of NaV1.6-Slac interaction on the cell surface.*

We thank the reviewer for good suggestions. We acknowledge that employing proximity ligation assay and high-resolution techniques would significantly enhance our understanding of the localization of the Slack-NaV1.6 coupling.

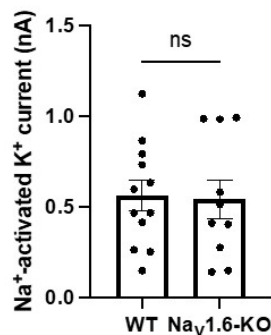
At present, the technical capabilities available in our laboratory and institution do not support high-resolution testing. However, we are enthusiastic about exploring potential collaborations to address these questions in the future. Furthermore, we fully recognize the importance of conducting co-immunoprecipitation (Co-IP) assays from membrane fractions. While we have already completed Co-IP assays for total protein and quantified the FRET efficiency values between Slack and NaV1.6 in the membrane region, the Co-IP assays on membrane fractions will be conducted in our future investigations.

*(2) Although hippocampal slices from Scn8a<sup>+/-</sup> were used for studies in Fig. S8, it is not clear whether Scn8a<sup>-/-</sup> or Scn8a<sup>+/-</sup> tissue was used in other studies (Fig 1J & 1K). It will be important to clarify whether genetic manipulation of NaV1.6 expression (Fig. 1K) has an impact on sodium-activated potassium current, level of surface Slack expression, or that of NaV1.6 near Slack.*

We thank the reviewer for pointing this out. In Fig. 1G,J,K, primary cortical neurons from homozygous NaV1.6 knockout (Scn8a<sup>-/-</sup>) mice were used. We will clarify this information in the revised manuscript. In terms of the effects of genetic manipulation of NaV1.6 expression on IKNa and surface Slack expression, we compared the amplitudes of IKNa measured from homozygous NaV1.6 knockout (NaV1.6-KO) neurons and wild-type (WT) neurons. The results showed that homozygous knockout of NaV1.6 does not alter the amplitudes of IKNa (Author response image 2). The level of surface Slack expression will be tested further.

### Author response image 2.

The amplitudes of IKNa in WT and NaV1.6-KO neurons (data from manuscript Fig. 1K). ns,  $p > 0.05$ , unpaired two-tailed Student's *t* test.

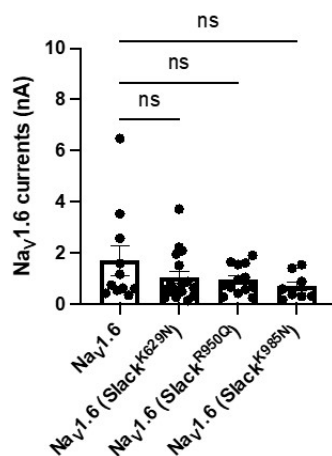


(3) Did the epilepsy-related Slack mutations have an impact on NaV1.6-mediated sodium current?

We thank the reviewer's question. We examined the amplitudes of NaV1.6 sodium current upon expression alone or co-expression of NaV1.6 with epilepsy-related Slack mutations (K629N, R950Q, K985N). The results showed that the tested epilepsy-related Slack mutations do not alter the amplitudes of NaV1.6 sodium current (Author response image 3).

### Author response image 3.

The amplitudes of NaV1.6 sodium currents upon co-expression of NaV1.6 with epilepsy-related Slack mutant variants (SlackK629N, SlackR950Q, and SlackK985N). ns,  $p > 0.05$ , one-way ANOVA followed by Bonferroni's post hoc test.



1. Showing the impact of quinidine on persistent sodium current in neurons and on NaV1.6-expressing cells would further increase confidence in the role of persistent sodium current on sensitivity of Slack to quinidine.

We appreciate the reviewer's question. Previous studies have shown that quinidine can inhibit persistent sodium currents at low concentrations<sup>1</sup>. In our experiments, blocking persistent sodium currents by application of riluzole in the bath solution showed no significant effects on the sensitivity of Slack to quinidine blockade upon co-expression of Slack with NaV1.6 (Fig. 2F,H). This result suggested that persistent sodium currents were not involved in the sensitization of Slack to quinidine blockade.

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